

Galois Theory

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Note: text in **red** means that an addition or correction was made in the notes.

1 Overview and References

Galois theory is the study of field extensions via automorphism groups. In other words, we study inclusions $K \subseteq L$ of fields, in terms of the group $G := \text{Aut}(L, K) := \{\varphi : L \rightarrow L \mid \varphi|_K = I_K\}$ of automorphisms of L fixing K pointwise. Under certain conditions, this leads to an amazing, order-reversing bijection between subfields of G and intermediate fields $K \subseteq E \subseteq L$ of the extension. Moreover, in the usual case where L is a finite-dimensional vector space over K , the group G is finite, of size at most equal to this dimension!

1.1 Roots of Polynomials

A main special case of this is where L is obtained from K by adjoining all the roots of a polynomial $f(x) \in K[x]$. In this case, the group G can be seen to be a subgroup of the group of permutations of the roots of f . If the roots are distinct, this group is S_d where d equals the degree of f . Famous applications of this case include:

- Unsolvability of the quintic: although there are general formulas for the solutions of polynomial equations in degree at most four, which use only radicals (and field oper-

ations), the Abel–Ruffini theorem states that this is *impossible* for general polynomial equations in degree at least five.

The reason for this is that a polynomial equation $f(x)$ is solvable using only radicals if and only if the group $G < S_d$ is solvable. Since all subgroups of S_4 are solvable, but not of S_5 (e.g., A_5 is simple and nonabelian), we obtain the theorem.

- The fundamental theorem of algebra: This states that every polynomial with complex coefficients has a complex root. To see this we use the extension $K = \mathbb{R} \subseteq \mathbb{C} = L$ with automorphism group S_2 , whose nontrivial element is complex conjugation. In other words, $\mathbb{R}$ is the field of elements fixed under complex conjugation. Galois theory then reduces the theorem to the statement that an *odd-degree* polynomial with real coefficients must have a real root (which in turn is an easy consequence of calculus).
- A conceptual explanation of the discriminant: recall that for a quadratic polynomial $f(x) = ax^2 + bx + c \in K[x]$, the discriminant is $D := b^2 - 4ac$. The roots of $f(x)$ are given by the quadratic formula, which is valued in $L = K[\sqrt{D}]$: that is, they are all in the field obtained by adjoining the square root of D . It turns out that the discriminant makes sense for polynomials of *every* degree; then $K[\sqrt{D}]$ no longer contains the roots of $f(x)$, but $K[\sqrt{D}]$ is a canonical subfield of the field L obtained by adjoining all the roots. Moreover, $\sqrt{D} \in K$ if and only if the automorphism group $\text{Aut}(L, K)$ is in the alternating group A_d (for f of degree d).
- A rigorous proof you cannot trisect an angle with a compass and straightedge: here basically the idea is that compass and straightedge constructions only allow you to take lengths you can already construct, and perform field operations (arithmetic) together with extracting square roots. Then one uses Galois theory to show that the lengths obtainable by trisecting an angle do not lie in a field given by adjoining iterated square roots: the latter would give a field extension of degree a power of two, whereas the former would have degree a multiple of three. Actually, this doesn't really require Galois theory, just multiplicativity of degree of extension fields (i.e., if M is an extension field over L , then $\dim_K M = \dim_K L \dim_L M$), which we will prove soon.

One of the main goals and motivation for the course will be to explain the above applications.

1.2 Relationships between fields and subfields

Actually, roots of polynomials, while historically the motivation for Galois theory, are not anymore the reason why Galois theory is so important and ubiquitous in modern mathematics (and other areas of science). Really, Galois theory is one of the most important examples of a correspondence between two a priori unrelated areas of mathematics. In this case, it is between *fields* and *field extensions* on the one hand, and *groups* and *subgroups* on the other hand. While the former are somewhat difficult to handle (fields being objects with two operations with somewhat complicated behaviour), groups can be much easier and more rigid. Moreover we will be interested in finite degree extensions: where L is a finite-dimensional

vector space over the subfield K . In this case the group G will turn out to be at most of size equal to this dimension, so we transform infinite and complicated fields into the study of a finite group, at least for the purpose of understanding the intermediate fields.

For example, this allows us to translate linear algebra over the larger field L into linear algebra over the smaller field K , by passing to elements invariant under the group G .

Consider the case $L = \mathbb{C}$ and $K = \mathbb{R}$, the subfield of conjugation-invariant elements. Every complex square matrix has a complex eigenvalue, but not every real square matrix has a real eigenvalue. Using complex conjugation, we see that every real matrix has complex eigenvalues occurring in conjugate pairs. Going deeper, over $\mathbb{C}$, every square matrix has a Jordan canonical form, which does not exist over $\mathbb{R}$. But, if you have a real matrix, thought of as a complex matrix it does have a Jordan form, and not just the eigenvalues, but the Jordan blocks themselves must appear in conjugate pairs. Using this observation we can get a corresponding canonical form over the reals, the only difference with Jordan form being that we need to look at block matrices with either 1×1 matrices on the diagonal (the real eigenvalues) or 2×2 matrices (matrices whose eigenvalues are a pair of complex conjugates).

Another example would be group representation theory—ways of realising groups as groups of matrices. This theory is simpler for $\mathbb{C}$ because it is algebraically closed, but Galois theory allows one to deduce representation theory over $\mathbb{R}$ as well (again using complex conjugation). In this sense, one can understand $\mathbb{R}$ as the conjugate-invariant complex numbers, which is a very special case of Galois theory.

Perhaps this all seems a bit funny, because $\mathbb{R}$ is defined first and $\mathbb{C}$ seems more complicated, but actually in linear algebra the opposite turns out to be true, and Galois theory allows you to deduce linear algebra over $\mathbb{R}$!

In the case of the extension $\mathbb{C}$ over $\mathbb{R}$, everything is quite simple as the group $\text{Aut}(\mathbb{C}, \mathbb{R}) \cong C_2$, with nontrivial element the complex conjugation—in other words, $\mathbb{R}$ is the subfield of $\mathbb{C}$ of numbers equal to their own conjugate. To deal with only this case, one could just talk more simply just about complex conjugation, and not develop Galois theory at all. But to deal with more complicated extensions (notably, the ones that arise in number theory, arithmetic geometry, or over fields of characteristic p), Galois theory is essential.

1.3 Uses of Galois theory in modern mathematics

Deeper examples include:

- Finite fields and the Frobenius morphism: $\mathbb{F}_p$ is the fixed locus under $\mathbb{F}_q$ of the Frobenius morphism $x \mapsto x^p$, where q is a power of p and p is prime (in fact this morphism generates $\text{Aut}(\mathbb{F}_q, \mathbb{F}_p)$);
- The study of number fields: much of number theory concerns the study of finite extension fields of the rational numbers. For instance, the Gaussian field $\mathbb{Q}[i]$ is quite important, and its study explains why the odd primes that are equal to the sum of two squares are the ones that are congruent to one modulo four.

- The Langlands program: this is a vast program relating number theory, analysis, and representation theory. Put simply, this studies the automorphism group $G := \text{Aut}(\overline{\mathbb{Q}}, \mathbb{Q})$ of the algebraic closure of $\mathbb{Q}$ via its representations: homomorphisms $G \rightarrow \text{GL}_n(\mathbb{C})$ to the group of $n \times n$ invertible complex matrices. It gives a correspondence between n -dimensional Galois representations and “automorphic representations” of the adelic group $\text{GL}_n(\mathbb{A}_{\mathbb{Q}})$ (we won’t explain what these are). The case $n = 1$ is the celebrated class field theory (which is itself a vast generalisation of Gauss’s quadratic reciprocity theorem); the case $n = 2$ includes the proof of Fermat’s Last Theorem.

We won’t be able to discuss much of the last two applications, but we will talk about the Frobenius morphism and Galois groups of extensions of finite fields.

1.4 References

I have loosely followed the lecture notes of Alessio Corti, *Galois Theory*. I deviated by adding content on angle trisection and on the fundamental theorem of algebra, and by not actually stating the axioms motivated by Grothendieck’s approach; I also added various other details from time to time.

There are actually gazillions of references that treat Galois theory or, what is quite similar, field theory. Here is a partial list of recommended references. If you find another one you like, let me know.

- E Artin, *Galois Theory*, University of Notre Dame.
- I N Herstein, *Topics in Algebra*, Xerox College Publishing, Lexington, Mass.-Toronto, Ont., second edition, 1975.
- Fraleigh, *A first course in abstract algebra*.
- I T Adamson, *Introduction to Field Theory*, Oliver and Boyd.
- D J H Garling, *A course in Galois theory*, CUP.
- I N Stewart, *Galois Theory*, Chapman and Hall.
- B L van der Waerden, *Algebra (or Modern algebra)*, vol. 1
- S Lang, *Algebra*, Springer
- I R Shafarevich, *Basic notions of algebra*, Springer
- J-P Tignol, *Galois’ theory of algebraic equations*, World scientific

2 Prerequisites (not to be covered in lectures)

The following is material we will use freely, and I won't cover it, unless you request it. If you aren't familiar with anything, please do read up on it and/or come and see me. I may add more material here in the future.

2.1 Linear algebra

You should be familiar with the basics of linear algebra over arbitrary fields, including linear independence, spanning sets, bases, dimension, eigenvectors and eigenvalues, matrices, change-of-basis matrices, traces, determinants, and characteristic polynomials. Here are some brief reminders, which are not exhaustive, and not a substitute for actually learning this material.

A field is a set with two commutative associative operations, $+$ and $\times$, called addition and multiplication, and *distinct* units 0 and 1 for addition and multiplication, such that every number has an additive inverse and every nonzero number has a multiplicative inverse, and such that multiplication distributes over addition: $x(y + z) = xy + xz$. A homomorphism of fields is a map $E \rightarrow F$ of fields which is additive and multiplicative and sends 1 to 1 (it also must send 0 to 0, although this is automatic).

A vector space V over a field K is an abelian group with a left multiplication by the field, called scalar multiplication, which satisfies associativity and distributivity and unit axioms; alternatively, we could encode all this structure into a ring homomorphism $\rho : K \rightarrow \text{End}_{\text{ab}}(V)$ where $\text{End}_{\text{ab}}(V)$ denotes the abelian group endomorphisms, i.e., self homomorphisms, of V , with the correspondence $\lambda \cdot x = \rho(\lambda)(x)$.

A subset $S \subseteq V$ spans if every element of V can be written as $\sum_{i=1}^n \lambda_i s_i$ for some $s_i \in S, \lambda_i \in K$ and some $n \geq 1$. Similarly, a tuple of elements $(s_1, \dots, s_n)$ spans V if every element of V can be written as $\sum_{i=1}^n \lambda_i s_i$ for some $\lambda_i \in K$. A subset $S \subseteq V$ is *linearly independent* if whenever $\sum_{i=1}^n \lambda_i s_i = 0$ for distinct $s_1, \dots, s_n \in S$, then $\lambda_i = 0$ for all i . Similarly, a tuple $(s_1, \dots, s_n)$ is linearly independent if $\sum_{i=1}^n \lambda_i s_i = 0$ implies $\lambda_i = 0$ for all i . (We remark that, unlike for spanning, linear independence of a set S and of a tuple $(s_1, \dots, s_n)$ are only equivalent when all the s_i are distinct: e.g., for a nonzero vector v , $S = \{v, v\} = \{v\}$ is linearly independent but not (v, v) .) A tuple or set which spans and is linearly independent is called a *basis*. Equivalently, $(s_1, \dots, s_n)$ forms a basis if every element can uniquely be written as a linear combination $a_1 s_1 + \dots + a_n s_n, a_i \in K$.

A fundamental result from linear algebra is that every vector space admits a basis. The size (cardinality) of the basis does not depend on the choice of basis, and is called the *dimension* of the vector space. We will only need this fact in the case of finite-dimensional vector spaces: we are then saying that if a vector space is spanned by finitely many vectors, then there exists a finite basis, and any two bases have the same size, which we call the dimension.

Linear algebra is the study of vector spaces and their linear transformations. A linear transformation $T : V \rightarrow W$ of K -vector spaces is a homomorphism of abelian groups which commutes with scalar multiplication: $T(\lambda v) = \lambda T(v)$ for all $v \in V$ and $\lambda \in K$. Given bases

$\mathcal{B} = (v_1, \dots, v_m)$ and $\mathcal{C} = (w_1, \dots, w_n)$ of V, W , we can write every linear transformation T as a matrix $[T]_{\mathcal{B}, \mathcal{C}}$, whose entries t_{ij} are such that $T(v_i) = \sum_j t_{ji} w_j$. For linear transformations with $V = W$, we usually require that both bases be the same and then write $[T]_{\mathcal{B}}$ (but this is not strictly necessary). The fundamental rule connecting matrix multiplication with composition of linear transformations is: $[T]_{\mathcal{B}, \mathcal{C}}[S]_{\mathcal{A}, \mathcal{B}} = [T \circ S]_{\mathcal{A}, \mathcal{C}}$. From this follows the change of basis formula for matrices: for $P\mathcal{B} = \mathcal{B}'$ we have $P = [I]_{\mathcal{B}', \mathcal{B}}$, hence $[T]_{\mathcal{B}'} = [I]_{\mathcal{B}, \mathcal{B}'}[T]_{\mathcal{B}}[I]_{\mathcal{B}', \mathcal{B}} = P^{-1}[T]_{\mathcal{B}}P$. In the case $V = W$, eigenvalues and eigenvectors of T (i.e., $Tv = \lambda v$ for some $v \in V$ and $\lambda \in K$) map to eigenvalues and eigenvectors of $[T]_{\mathcal{B}}$, so we see that conjugate matrices have the same eigenvalues (a fact which can also be easily seen directly).

The trace of a square matrix is the sum of the diagonal entries. The trace of a linear transformation $T : V \rightarrow V$ is the trace of a matrix $[T]_{\mathcal{B}}$. This does not depend on the choice of basis $\mathcal{B}$ because traces of matrices satisfy $\text{tr}(AB) = \text{tr}(BA)$, and hence $\text{tr}(P^{-1}AP) = \text{tr}(PP^{-1}A) = \text{tr}(A)$.

The determinant of a linear transformation or matrix equals the product of its eigenvalues with (algebraic) multiplicity. That is, there exists a basis $\mathcal{B}$ in which $[T]_{\mathcal{B}}$ is upper-triangular, with eigenvalues with multiplicity being the diagonal entries, so $\det(T)$ equals the product of the diagonal entries of $[T]_{\mathcal{B}}$ where the latter is upper triangular (this does not depend on the choice of $\mathcal{B}$ because we can define eigenvalues and even their multiplicity without bases: the multiplicity of an eigenvalue λ is the maximum value of $\dim \ker(T - \lambda I)^m$ for $m \geq 1$). The determinant of a square matrix is also given by an explicit formula $\det(A) = \sum_{\sigma \in S_n} \text{sign}(\sigma) a_{i\sigma(i)}$, where $\text{sign} : S_n \rightarrow \{\pm 1\}$ is the unique group homomorphism sending all transpositions to -1 . The determinant of a matrix is zero unless it is invertible. We have the formulas $\det(I) = 1$ and $\det(AB) = \det(A)\det(B)$, so that restricted to the invertible $n \times n$ matrices, denoted $\text{GL}_n(K)$, the determinant defines a group homomorphism $\text{GL}_n(K) \rightarrow K^\times := K \setminus \{0\}$. The determinant of a matrix equals the product of its eigenvalues with multiplicity. In particular, the determinant of an upper triangular matrix is the product of the diagonal entries. The determinant of a linear transformation $T : V \rightarrow V$ satisfied $\det T = \det[T]_{\mathcal{B}}$ for any basis $\mathcal{B}$ (the choice of basis does not matter due to the multiplicativity of the determinant). There are other more sophisticated ways to view the determinant. For example, there is a unique skew-symmetric multilinear map $\alpha : V^{\dim V} \rightarrow K$ (multilinear means linear in each component) up to scaling; then we could define the determinant by the formula $\alpha \circ (A^{\dim V}) = \det(T)\alpha$, with the left hand side meaning $(v_1, \dots, v_{\dim V}) \mapsto \alpha(Av_1, \dots, Av_{\dim V})$, which is still skew-symmetric and multilinear. If you know about exterior product of vector spaces, this can alternatively be written as $\det T = \wedge^{\dim V} T$.

2.2 Ring theory

You also should have a basic background on rings, including: rings, ring homomorphisms, ideals, prime and maximal ideals, integral domains, Euclidean domains, principal ideal domains (PIDs), unique factorisation domains (UFDs), Gauss's Lemma for polynomial rings in one variable over a UFD. Briefly, a ring R is an abelian group with a distributive associative

multiplication having a identity $1 \in R$. A ring homomorphism is an additive multiplicative map $R \rightarrow S$ sending 1 to 1 (also 0 to 0, but the latter is automatic). A subring is a subset containing 1 and closed under addition, multiplication, and negation (hence also containing 0). An ideal is an abelian subgroup $I \subseteq R$ closed under multiplication by R (for commutative rings, we just say $aI \subseteq I$ for all $a \in R$; for noncommutative rings, these are *left ideals*, and one has also right ideals defined by $Ia \subseteq I, \forall a \in R$, and two-sided ideals, which are simultaneously left and right ideals). The kernel of a ring homomorphism $\varphi : R \rightarrow S$ is an ideal, and the first isomorphism theorem gives $\bar{\varphi} : R/I \rightarrow S$, an injective homomorphism (hence an isomorphism onto its image). A maximal ideal is a proper ideal not contained in any other proper ideal. A prime ideal P is one such that $ab \in P$ implies either $a \in P$ or $b \in P$. An integral domain is a commutative ring in which $ab = 0$ implies either $a = 0$ or $b = 0$. Equivalently, these are commutative rings where the zero ideal is prime. More generally, a quotient R/I of a commutative ring is an integral domain if and only if I is prime. A field is a commutative ring in which every nonzero element is multiplicatively invertible. Equivalently, this is a commutative ring F in which 0 and F are the only ideals. In particular, fields are integral domains. A quotient R/I of a commutative ring is a field if and only if I is maximal. In particular, every maximal ideal is prime, but not conversely. A Euclidean domain is an integral domain R equipped with a valuation $\nu : R \rightarrow \mathbb{N}$ such that one can perform the Euclidean algorithm: for every $a, b \in R$, there exists $q, r \in R$ with $\nu(r) < \nu(b)$ such that $a = bq + r$. A PID is an integral domain in which every ideal is generated by a single element (such an ideal is called principal). A unique factorisation domain is one in which every element can be uniquely factored into irreducible elements, the latter meaning nonzero elements a which are not units such that $a = bc$ implies that either b or c is a unit. We have the implications: Euclidean domain implies PID implies UFD; the first is obtained by applying the Euclidean algorithm, and the composite implication can also be obtained using the Euclidean algorithm. The direct implication that PID implies UFD is a little more technical, but one way is to first observe that PIDs are Noetherian: every ascending chain $I_1 \subseteq I_2 \subseteq \cdots$ stabilises, i.e., has $I_n = I_{n+1} = \cdots$ for some n , since the union $\bigcup I_i$ is an ideal whose generator must be contained in some I_n . Then, one can show that a Noetherian integral domain is a UFD if and only if every irreducible element a is prime, i.e., the ideal (a) is prime. (Note that there are UFDs which are not Noetherian, such as $\mathbb{C}[x_1, x_2, x_3, \dots]$, although in this case it is still true that every irreducible element is prime, and one has the ascending chain condition for principal ideals but not all ideals.)

Gauss's Lemma is as follows: Let R be a UFD. Then, if both of $f, g \in R[x]$ have coprime coefficients, so does fg . As a consequence, for K the field of fractions of R , if $f \in R[x]$ is irreducible, then it is also irreducible in $K[x]$. This is because, if $f = gh$ for $g, h \in K[x]$, then after clearing denominators, we are left with the equality $cf = \tilde{g}\tilde{h}$ where $\tilde{g}, \tilde{h} \in R[x]$ have coprime coefficients. Then it follows from Gauss's Lemma that $c \in R^\times$, so f is reducible as well, a contradiction. Note that, conversely, if $f \in R[x]$ has coprime coefficients and is irreducible in $K[x]$, it is clearly irreducible in $R[x]$. So we get the biconditional: irreducibility for $f \in R[x]$ in $R[x]$ is equivalent to irreducibility in $K[x]$ together with having coprime coefficients. Put together with the fact that $K[x]$ is a UFD (since it is a Euclidean domain,

as above), we conclude also that $R[x]$ is a UFD.

Another useful way to think about Gauss's Lemma is as follows: if $f \in \mathbb{Q}[x]$ is nonzero, there is a unique rational number $q \in \mathbb{Q}$ up to sign such that $q^{-1}f$ has coprime integral coefficients. We call q the *content* of f , $\text{cont}(f)$. Gauss's Lemma says that $\text{cont}(f) = 1 = \text{cont}(g)$ implies $\text{cont}(fg) = 1$. This can be restated, by multiplying by scalars, as

$$\text{cont}(fg) = \text{cont}(f)\text{cont}(g), \forall f, g \in \mathbb{Q}[x].$$

In particular, this implies that, if fg and f are in $\mathbb{Z}[x]$ with coprime coefficients, and $g \in \mathbb{Q}[x]$, then g is also in $\mathbb{Z}[x]$ with coprime coefficients. (Explicitly, there is $q \in \mathbb{Q}$ such that $qg \in \mathbb{Z}[x]$ with coprime coefficients, then $qfg = f(qg)$, so by Gauss's Lemmas, $qfg \in \mathbb{Z}[x]$ with coprime coefficients, so $q = \pm 1$.)

3 Basics on fields

In a sense, a course on Galois theory is really a study of fields, more precisely, of finite extensions of fields (and especially the separable and normal ones). Fundamentally we have to learn some basic theory of fields. Then, thanks to the main theorem of Galois theory, we can translate questions about these extensions to questions about finite groups, which is extremely powerful and gives lots of information about the fields. Although the philosophy of Galois theory also extends to many similar situations, such as covering spaces in topology (and there is even a common framework unifying field extensions with covering spaces in the context of algebraic geometry or complex manifolds, so that the Galois group literally becomes the fundamental group), in this course we will be uniquely concerned with fields. In this section we will cover the basics that we need on fields.

3.1 Injectivity of field homomorphisms

The following result, although easy, is quite fundamental and useful, and special to the case of fields:

Lemma 3.1.1. Every homomorphism of fields $\alpha : K \rightarrow L$ is injective.

Proof. If $\alpha(x) = \alpha(y)$, then $\alpha(x - y) = 0$. If $x \neq y$, then there exists p with $p(x - y) = 1$, so that $\alpha(1) = \alpha(p(x - y)) = \alpha(p)0 = 0$. This contradicts an axiom. $\square$

Remark 3.1.2. Here it is essential that we included as an axiom of a homomorphism that 1 be mapped to 1. Had we not required this, the map sending everything to zero would be a homomorphism, which is clearly not injective.

Corollary 3.1.3. Every homomorphism of fields $\alpha : K \rightarrow L$ is an isomorphism onto its image, a subfield of L .

Proof. Call $K' := \alpha(K)$ the image, clearly a subfield. Then the induced map $\alpha' : K \rightarrow K'$ is a bijection, and the inverse map is also a field homomorphism, so it is an isomorphism. $\square$

Definition 3.1.4. Given fields K, L , we denote by $\text{Emb}(K, L)$ the set of field homomorphisms K to L . We call these embeddings since they are injective.

3.2 Field extensions as vector spaces and degree

Given a field extension $K \subseteq L$, L is equipped with the structure of a K -vector space (in fact, a K -algebra).

Definition 3.2.1. The degree of the extension, denoted $[L : K]$, is defined as the dimension, $\dim_K L$, of L as a K -vector space. The extension is called *finite* if the degree is finite.

Example 3.2.2. We have $[\mathbb{C} : \mathbb{R}] = 2 = [\mathbb{Q}[i] : \mathbb{Q}]$. More generally, if K is a field and $L = K[\sqrt{a}]$ is the ring obtained by adjoining the square root of an element $a \in K$ which is not a square in K , then a basis for L over K is $1, \sqrt{a}$. One can easily check that L is a field, since $(b + c\sqrt{a})(b - c\sqrt{a}) = b^2 - ac^2$, which is nonzero for $b, c \in K$ as a is not a square. So L is a field, and $[L : K] = 2$. Such extensions are called *quadratic*.

3.3 The tower law

Proposition 3.3.1. Given a chain of finite extensions $K \subseteq L \subseteq M$, we have $[M : K] = [M : L][L : K]$.

Proof. Let $b_1, \dots, b_m$ be a basis for L over K and let $c_1, \dots, c_n$ be a basis for M over L . We claim that $\mathcal{B} := (c_i b_j)_{1 \leq i \leq m, 1 \leq j \leq n}$ is a basis for M over K . Indeed, any element of M can be uniquely written as an L -linear combination of the c_i , and each coefficient can be uniquely written as a K -linear combination of the b_j . So every element of M can uniquely be written as a K -linear combination of elements of $\mathcal{B}$, and the latter is a basis. Its size is mn . $\square$

Example 3.3.2. One can check that 3 is not a square in $\mathbb{Q}[i]$. So $[\mathbb{Q}[i, \sqrt{3}] : \mathbb{Q}] = [\mathbb{Q}[i][\sqrt{3}] : \mathbb{Q}[i]][\mathbb{Q}[i] : \mathbb{Q}] = 4$. By the fundamental theorem of Galois theory we will prove later, the only intermediate fields between $\mathbb{Q}[i, \sqrt{3}]$ and $\mathbb{Q}$, aside from these themselves, are $\mathbb{Q}[i]$, $\mathbb{Q}[i\sqrt{3}]$, and $\mathbb{Q}[\sqrt{3}]$. This can also be checked directly in this case.

Example 3.3.3. Using this we can prove that it is impossible to trisect an angle with a compass and straightedge. First, the basic theory of the compass and straightedge: using these, given a set of constructible lengths, we can perform field operations (addition, multiplication, division, subtraction), as you probably have seen (one uses similar triangles for multiplication and division). Beginning with a single length, call it 1, then we can construct all rational lengths $\mathbb{Q}$. By performing further constructions we can always obtain a field $K \supseteq \mathbb{Q}$. The constructions that can be performed are:

- Intersecting lines defined over K : the intersection point is still defined over K (it is in $K^2 \subseteq \mathbb{R}^2$).
- Intersecting a line and a circle: this produces a quadratic equation, so lies in a possible quadratic extension $L^2 \subseteq \mathbb{R}^2$ (either $L = K$ or $[L : K] = 2$).
- Intersecting two circles: this also produces a single quadratic equation, so it is the same as the previous point.

- Copying the length between two constructed points onto the x axis: this involves only field operations and a square root, so again the field K of constructed lengths can only be extended quadratically.

Any length obtained by finitely many compass and straightedge operations therefore lies in a field obtained by finitely many iterated quadratic extensions of $\mathbb{Q}$. By the tower law, such a field has degree a power of two.

On the other hand, if we can trisect the angle $\pi/3 = 60^\circ$, we can construct $\cos(\pi/9)$, by drawing a right triangle with hypotenuse 1 and angle $\pi/9$. By the triple angle formula, $\cos(3\theta) = 4\cos^3(\theta) - 3\cos(\theta)$, we have that $\cos(\pi/9)$ satisfies the polynomial $4x^3 - 3x - \frac{1}{2}$, hence also $8x^3 - 6x - 1$. By the rational root theorem (a special case of Gauss's Lemma, which we will recall later), the only possible rational roots have numerator dividing the constant term, -1 , and denominator dividing the leading term, 8 , so they could only be $\pm 1, \pm \frac{1}{2}, \pm \frac{1}{4}, \pm \frac{1}{8}$. We can check these do not satisfy $8x^3 - 6x - 1$, so that $8x^3 - 6x - 1$ does not have any rational linear factors and hence is irreducible. As we explain in the next subsection, this implies that the field generated by $\cos(\pi/9)$ has degree three. Since this is not a factor of a power of two, $\cos(\pi/9)$ cannot be constructed.

3.4 Minimal polynomials

To give a more general example, it will be important to understand the notion of *minimal polynomial*. Given an extension of fields $K \subseteq L$, and an element $b \in L$, we can form the polynomial ring $K[x]$, and a homomorphism $K[x] \rightarrow L$ sending x to $b \in L$. If L is a finite degree extension, this map cannot be injective, as it is a K -linear transformation and $K[x]$ is infinite dimensional over K . The kernel is then a nonzero ideal in $K[x]$. A basic fact is that this is generated by a single element $f(x)$:

Definition 3.4.1. Given an extension $K \subseteq L$ of fields, and an element $b \in L$, the minimal polynomial is the unique monic polynomial $f(x) \in K[x]$ such that $(f(x)) = \ker(K[x] \rightarrow L, x \mapsto b)$.

Let's recall a bit more about the aforementioned property of $K[x]$. Recall that we can do long division on polynomials: given polynomials $f(x), g(x)$, there exists unique polynomials $q(x)$ and $r(x)$ with $\deg r(x) < \deg f(x)$ such that $g(x) = f(x)q(x) + r(x)$. This allows one to show that any nonzero element of an ideal of $K[x]$ of minimal degree is a generator. (Formally, we have shown that $K[x]$ is a Euclidean domain, and every Euclidean domain is a principal ideal domain (PID), meaning an integral domain where every ideal is generated by a single element.) Moreover, this shows us that if $f(x) \in K[x]$ is an irreducible polynomial, then $(f(x))$ is a maximal ideal: indeed, if I is an ideal properly containing it, then $I = (g(x))$, and $f(x) \in I$ implies $g(x) \mid f(x)$, but irreducibility of f then implies that $g(x)$ is a constant, so I is the unit ideal.

Returning to the situation of the kernel $(f(x))$ of $K[x] \rightarrow L$, we can apply the first isomorphism theorem to obtain an injective ring homomorphism $K[x]/(f(x)) \rightarrow L$.

Proposition 3.4.2. The image of this map is a field, which is an extension of K with basis $1, b, b^2, \dots, b^{\deg f - 1}$, of degree $\deg f$. Moreover, $f(x)$ is the unique monic irreducible polynomial that b satisfies.

Proof. Call the image $K[b] \subseteq L$. First of all, as a subring of a field, $K[b]$ is an integral domain. Thus, the kernel $(f(x))$ is a prime ideal. As a result, $f(x)$ is an irreducible polynomial. Then, the ideal $(f(x))$ is maximal. So the quotient $K[x]/(f(x)) \cong K[b]$ is a field. Finally, by using the Euclidean algorithm, every polynomial $g(x) \in K[x]$ can be written uniquely modulo $f(x)$ as a polynomial of degree $< \deg f(x)$. So the basis is as described. Finally, we showed that $f(x)$ is irreducible. Conversely, any polynomial which b satisfies must be in the ideal $(f(x))$, and to be monic and irreducible it has to be $f(x)$. $\square$

Example 3.4.3. If L is a field extension of K generated by a single element $b \in L$ not in K , the above implies that $[L : K] = \deg f(x)$, for $f(x)$ the minimal polynomial.

Example 3.4.4. Let ζ be a primitive fifth root of unity. Then ζ satisfies an irreducible polynomial $x^4 + x^3 + x^2 + x + 1$ (as an exercise you can prove that it is irreducible, using Gauss's Lemma!) So $(\mathbb{Q}[\zeta] : \mathbb{Q}) = 4$.

Remark 3.4.5. Here is a solution to the exercise: by Gauss's Lemma it is enough to show irreducibility over the integers. If there were a linear factor $ax + b$ then $a \mid 1, b \mid 1$, so we would have $x \pm 1$ is a factor, which it is clearly not as ± 1 are not roots. So we need to look for a quadratic factor $ax^2 + bx + c$, and again it is enough to assume $a = 1, c = \pm 1$. The other factor would have to be of the same form, so we would have a product like this:

$$(x^2 + bx \pm 1)(x^2 + cx \pm 1) = x^4 + (b + c)x^3 + (bc \pm 2)x^2 \pm (b + c)x + 1.$$

We see that for this to give us the desired polynomial we need to simultaneously solve:

$$b + c = \pm(b + c) = (bc \pm 2) = 1,$$

and this can only happen if the signs are plus, $b + c = 1$, and also $bc + 2 = 1$. Substituting $c = 1 - b$ the last equation becomes $b(1 - b) + 1 = 0$, i.e., $b^2 - b - 1 = 0$, but this does not have any integral solutions. So we get irreducibility.

As pointed out by a student in lecture, a slicker, but less straightforward, way to prove irreducibility is to make the substitution $x \mapsto y + 1$ and get the polynomial $y^4 + 5y^3 + 10y^2 + 10y + 5$, then to note that this polynomial is irreducible by Eisenstein's criterion (all non-leading coefficients are multiples of the prime number 5, but the constant term is not a multiple of 25). (Here is a quick proof of Eisenstein's criterion: if p divides all non-leading terms than any integral factorisation reduces modulo p to a product of powers of y . So an integral factorisation would be into factors of the form $y^a + pg$ for $\deg g < a$. A nontrivial such factorisation has constant term a multiple of p^2 .)

3.5 Splitting field extensions

Given a field extension $K \subseteq L$, an *intermediate field* is a field $K' \subseteq L$ containing K .

Definition 3.5.1. A finite field extension L of K is called a *splitting field* of the polynomial $f(x) \in K[x]$ if $f(x)$ splits into linear factors in $L[x]$ and this property is not satisfied for any other intermediate field K' .

Equivalently, L is a splitting field of f if f splits into linear factors and L is generated as a field by its roots.

Proposition 3.5.2. Given a field K and a polynomial $f \in K[x]$, there exists a splitting field L of f over K .

Proof. We prove this by induction on the degree of f . We can assume that f has no linear factors, because the splitting field of f and $f/(x-a)$ is the same for every $a \in K$. Suppose now that $\deg f \neq 0$ and that $g \mid f$ is an irreducible factor with $\deg g > 1$. Form $K' := K[x]/(g(x))$, a field extension of K generated by a root of $g \mid f$. Over K' , f has a linear factor, so the splitting field of f over K' is the same as that of a polynomial of strictly lower degree. By the induction hypothesis, there exists a splitting field L of f over K' . Then L is generated by the roots of f over K' . Since K' is itself generated by a root of f over K , overall L is generated by roots of f over K . So it is a minimal field over which f factors into linear factors, hence a splitting field. $\square$

The importance of splitting field extensions is the following alternative definition:

Definition 3.5.3. An extension L of K is normal if any two K -linear field embeddings $\varphi_1, \varphi_2 : L \rightarrow M$ differ by precomposition with a K -linear automorphism of L .

In other words, given an embedding $\varphi : L \rightarrow M$, the map $\text{Aut}_K(L) \rightarrow \text{Emb}_K(L, M), \psi \mapsto \psi \circ \varphi$ is surjective. Equivalently, any two K -linear embeddings into M have the same image, because in this case the embedding can only differ by an automorphism of L . In particular, in this case, this image is invariant under all K -linear automorphisms of M .

Proposition 3.5.4. A splitting field extension is normal.

Proof. Suppose that L is a splitting field of a polynomial f , and let $\varphi : L \rightarrow M$ be a K -linear embedding, for $K \subseteq M$. Then the image $\varphi(L)$ is also a splitting field of f (as it is an isomorphic extension of K). This means however that $\varphi(L)$ equals the subfield of M generated by all roots of f . So it does not depend on the choice of φ . $\square$

Later on we will prove the **partial** converse that every **finite** normal extension is also a splitting field extension. **Thus, splitting field extensions are equivalent to finite normal extensions.**

Remark 3.5.5. Note that a normal extension need not be finite: an example of an infinite normal extension would be the extension $K \supseteq \mathbb{Q}$ spanned by all complex n -th roots of unity for all n . This is normal because any ($\mathbb{Q}$ -linear) embedding $K \rightarrow L$ has image generated (over $\mathbb{Q}$) by the n -th roots of unity in L . (I put round brackets here because every field of characteristic zero canonically contains $\mathbb{Q}$ by fractions of sums of 1 with itself along with negation, and every field homomorphism with source (equivalently target) of characteristic zero is automatically $\mathbb{Q}$ -linear.)

The term normal is relevant here because of the following result:

Proposition 3.5.6. Let $K \subseteq L \subseteq M$ be a sequence of field extensions, with L a normal extension of K . Then the subgroup $\text{Aut}_L(M) \leq \text{Aut}_K(M)$ is normal. Moreover, in this case there is a natural injective homomorphism $\text{Aut}_K(M)/\text{Aut}_L(M) \rightarrow \text{Aut}_K(L)$.

Remark 3.5.7. We will prove later that, if M is a normal extension of K , then the injective homomorphism above is in fact surjective, yielding an isomorphism $\text{Aut}_L(M)/\text{Aut}_K(M) \cong \text{Aut}_K(L)$.

Proof. It suffices to show that L is preserved by all K -linear automorphisms of M , for then a composition $\psi\varphi\psi^{-1}$ for $\psi \in \text{Aut}_K(M)$, $\varphi \in \text{Aut}_L(M)$, would act on L by $\psi|_L\psi^{-1}|_L = I_L$. But we already remarked above that this is true when L is normal. For the final statement, given a K -linear automorphism $\varphi : M \rightarrow M$, the restriction $\varphi|_L : L \rightarrow M$ must have image equal to L , since L is normal. So the restriction map $\varphi \mapsto \varphi|_L$ yields a well-defined map $\text{Aut}_K(M) \rightarrow \text{Aut}_K(L)$. This is clearly compatible with composition, so it is a homomorphism. The kernel consists of those automorphisms which are the identity on L , i.e., the L -linear ones. So we get an injective homomorphism $\text{Aut}_K(M)/\text{Aut}_L(M) \rightarrow \text{Aut}_K(L)$. $\square$

3.6 Separable field extensions

Definition 3.6.1. A polynomial which splits into linear factors is called *separable* if it has distinct roots. Otherwise it is called *inseparable*.

The following is a useful criterion to be separable:

Proposition 3.6.2. (Jacobian criterion) Let $f(x)$ be a polynomial splitting into linear factors. Then f is separable if and only if f and f' (the derivative) are coprime: $\gcd(f, f') = 1$.

Proof. Write $f = (x - a_1) \cdots (x - a_n)$. Then $f' = \sum_i \frac{f}{x - a_i}$. If $a_i = a_j$ for some $i \neq j$, then $(x - a_i)$ divides every term in this sum, so it divides $\gcd(f, f')$. If the a_i are all distinct, then for each i , $x - a_i$ divides all but one term in the sum, so it cannot divide f' . So in the latter case, none of the factors of f are factors of f' , hence $\gcd(f, f') = 1$. $\square$

Note now that if $f(x) \in K[x]$ does not split into linear factors, then the condition $\gcd(f, f') = 1$ still makes sense. The greatest common divisor does not change under field extension, so we immediately obtain the following:

Corollary 3.6.3. The following conditions are equivalent for $f \in K[x]$:

- $\gcd(f, f') = 1$;
- $f(x)$ is separable in any given field extension in which it splits into linear factors;
- $f(x)$ is separable in every such field extension.

Definition 3.6.4. Call such a polynomial $f \in K[x]$ separable, regardless whether it splits into linear factors. If it is not separable call it inseparable.

Definition 3.6.5. A field extension L of K is called separable if for every $b \in L$, its minimal polynomial $f(x)$ is separable. Otherwise it is called inseparable.

Note that, unlike in the normal/splitting field case, it follows that any intermediate extension $K_1 \subseteq K_2$ fitting into $K \subseteq K_1 \subseteq K_2 \subseteq L$ is *also* separable. Indeed, the minimal polynomial of an element $b \in K_2$ over K_1 is a factor of the minimal polynomial of b over K , and $b \in L$.

Remark 3.6.6. Later on we will prove the following facts:

- If $K \subseteq L \subseteq M$ and $K \subseteq L, L \subseteq M$ are both separable, then so is $K \subseteq M$;
- If $L = K(\alpha)$ for α having separable minimal polynomial, then L is a separable extension of K .

As a result, any field extension generated by elements with separable minimal polynomials is separable. Conversely, a separable extension is generated by elements with separable minimal polynomials. In particular, a splitting field of f is separable if and only if f is itself separable.

Remark 3.6.7. Also, later on, we will prove and use the following more sophisticated interpretation of separable extensions (analogous to the definition of normal extensions): An extension $K \subseteq L$ is separable if and only if, for every chain $K \subseteq K_1 \subsetneq K_2 \subseteq L$ of field extensions, there exists a field M and two distinct embeddings $\varphi_1, \varphi_2 : K_2 \rightarrow M$ which are equal on K_1 . This says that embeddings can “separate” K_2 from K_1 , in a sense (related to the fact that the extension of K_2 by K_1 is by roots of polynomials having distinct roots).

Proposition 3.6.8. Suppose that the characteristic of K does not divide n , and $f(x) \in K[x]$ is irreducible of degree n . Then f is separable.

As an immediate consequence, we have:

Corollary 3.6.9. If K has characteristic zero, then every finite field extension is separable.

Proof of Proposition 3.6.8. We have that $f'(x) = nx^{n-1} + \dots$ is a nonzero polynomial of degree $n - 1 < n$. So $f'(x) \notin (f(x)) \subseteq K[x]$. But since $f(x)$ is irreducible in K , the ideal $(f(x))$ is maximal (as we recalled earlier). So $(f(x), f'(x)) = (1)$, and $f(x)$ and $f'(x)$ are coprime. $\square$

We note that the converse to the proposition does not hold: for example, over $\mathbb{F}_2$, we can take the irreducible polynomial $x^2 + x + 1$, which is separable of degree two: indeed, $(x^2 + x + 1)' = 1$ so the Jacobian criterion holds. However we have the following stronger statement:

Proposition 3.6.10. An irreducible polynomial $f(x) \in K[x]$ is inseparable if and only if, for $p = \text{char } K$, we have $f(x) \in K[x^p]$. In this case, for some finite extension L of K obtained by adjoining p -th roots of some elements, we have $f(x) = g(x)^p$ for some polynomial $g \in L[x]$.

Proof. We see from the proof that $\gcd(f, f') = 1$ if and only if $f' \neq 0$. Note that $f' = 0$ is true if and only if every exponent of x appearing in f is a multiple of p . This proves the first statement. In this case, we could write $f(x) = \sum_i a_i x^{ip}$. Pass to a field extension in which, for every i , $a_i = b_i^p$ for some b_i . Then, by the “Freshers’s Dream”, realised in characteristic p , we have $(\sum_i b_i x^i)^p = \sum_i a_i x^{ip}$, so $f(x) = g(x)^p$ for $g(x) = \sum_i b_i x^i$. This proves the second statement. $\square$

Corollary 3.6.11. A field K has the property that, for every finite extension L and every $b \in L$, the minimal polynomial of b over K is separable, if and only if either:

- The characteristic of K is zero; or
- The characteristic of K is $p > 0$ and for every $a \in K$, there exists a p -th root $\sqrt[p]{a}$ of a in K .

Proof. It remains to consider the case of $\text{char } K = p > 0$. Suppose that every element of K admits a p -th root. If $b \in L$ has inseparable minimal polynomial $f(x)$, then by Proposition 3.6.10, since we have all the p -th roots in K already, we have $f(x) = g(x)^p$ for $g(x) \in K$, but that contradicts irreducibility. Conversely, suppose that $a \in K$ does not have a p -th root. Then $x^p - a$ is a polynomial which splits into linear factors over $K(b)$, $b^p = a$, as $(x - b)^p$. Since $b \notin K$ and $\text{char } K = p$, we see that $(x - b)^m \notin K[x]$ for $m < p$. Thus $x^p - a$ is irreducible, and as b satisfies it, it is the minimal polynomial of b . But it is inseparable. $\square$

Definition 3.6.12. A field is called *perfect* if every minimal polynomial of every element of every finite extension is separable.

By the corollary, a perfect field is a field either of characteristic zero, or of characteristic p with every element admitting a p -th root.

3.7 The fundamental theorem of Galois theory

Later on, we will define the notion of a Galois extension, and it will turn out to be equivalent to a separable splitting field extension. These will be the extensions $K \subseteq L$ for which we can obtain a bijection between the intermediate fields and the subgroups of the automorphism group $\text{Aut}_K(L)$.

Theorem 3.7.1 (The fundamental theorem of Galois theory—first version). Suppose that $K \subseteq L$ is a separable splitting field extension (or equivalently, finite normal separable extension). Then, the maps $H \rightarrow L^H, K' \mapsto \text{Aut}_{K'}(L)$ are inverse bijections between subgroups of $\text{Aut}_K(L)$ and subfields of L containing K . Moreover, this map sends normal subgroups to normal extensions of K and vice-versa, and for $K' \supseteq K$ a normal extension contained in K , we have an isomorphism $\text{Aut}_K(K') \cong \text{Aut}_K(L) / \text{Aut}_{K'}(L)$.

Definition 3.7.2. A Galois extension is a separable splitting field extension.

Remember by Remark 3.6.6, that a Galois extension is the same thing as a splitting field of a separable polynomial.

Note that it is an immediate consequence of the theorem statement that, if $H_1 < H_2$ are two subgroups of $\text{Aut}_K(L)$, then $L^{H_1} \supset L^{H_2}$, i.e., the bijection is order-reversing. We also remark that it is straightforward to show that if either of the maps in the theorem is a bijection, then the other is its inverse, so we could have given only one bijection.

Before we set about the task of proving this theorem, which will take at least a couple weeks, we will apply it to demonstrate how it yields a better understanding of field extensions.

3.8 Example: extensions by radicals

Let K be a field and $a \in K$. We will consider a field extension of the form $L = K(b)$ for $b^n = a$. Loosely, we may write $L = K(\sqrt[n]{a})$, although in practice this is imprecise notation as illustrated by this example:

Example 3.8.1. Let $K = \mathbb{Q}[\sqrt{2}]$. We can then consider $K[\sqrt[4]{2}]$. If we mean the positive fourth root (as is often the convention working inside $\mathbb{R}$), then this is the quadratic extension of K adjoining $\sqrt{\sqrt{2}}$. If we instead mean $i\sqrt{\sqrt{2}}$, then we get the quadratic extension of K adjoining $\sqrt{-\sqrt{2}}$. These two extensions are not isomorphic as extensions of K , since one splits the polynomial $x^2 - \sqrt{2}$ and the other splits $x^2 + \sqrt{2}$ (although, they are both isomorphic as extensions of $\mathbb{Q}$, since they adjoin a single root of $x^4 + 2$, and this polynomial is irreducible, hence both fields are isomorphic over $\mathbb{Q}$ to $\mathbb{Q}[x]/(x^4 + 2)$.)

In the above example, and for the remainder of these notes, we say that two fields L, L' containing a field K are isomorphic *over* K if there exists an isomorphism $L \rightarrow L'$ which is the identity on K .

Now back to the situation of $L = K(b)$ for $b^n = a$.

Question: Is L a splitting field extension? It is adjoining a root of $x^n - a$. But does it contain all of them?

Baby version: case $n=2$: In this case, the answer is yes: $x^2 - a = (x - \sqrt{a})(x + \sqrt{a})$, and $-\sqrt{a} \in L = K[\sqrt{a}]$.

Remark: Here we see a difference in the case that K has characteristic two: in this case, $\sqrt{a} = -\sqrt{a}$, so there is only one root of $x^2 - a$, and it is not separable. In all other cases, it is separable.

General case: The answer is no: if b, b' are roots of $x^n - a$, then b/b' is a root of $x^n - 1$, the n -th roots of unity. But we do not have to have all of these roots in $K(b)$. For example, if we take $K = \mathbb{Q}, a = 2, n = 3$, we could let $L = \mathbb{Q}(\sqrt[3]{2})$, taking the positive cube root. This is contained in $\mathbb{R}$, so cannot contain a primitive cube root of unity (as these are not in $\mathbb{R}$). So $\sqrt[3]{2}$ is the only root of $x^3 - 2$ in this field.

In order to give the general answer about extensions by radicals, an important simplification is to adjoin roots of unity. More precisely, we have the following facts.

Definition 3.8.2. For a field K , and $n \geq 1$, let $\mu_n(K) := \{\zeta \in K \mid \zeta^n = 1\}$ denote the group of n -th roots of unity in K .

Recall the following fact: every finite subgroup of a field is cyclic. (This is a good exercise, either to prove yourself or to read, but here is the reason why it holds: such a group is finite abelian, hence a product of cyclic groups, but for every k , there can be at most k elements of order k , as these are roots of $x^k - 1$; this can only happen if the group is actually cyclic.) So then $\mu_n(K) \cong C_m$ for some $m \mid n$ (the maximal order of an element of $\mu_n(K)$).

- Lemma 3.8.3.** 1. The polynomial $x^n - 1$ is separable if and only if the characteristic of K does not divide n .
2. Suppose that the characteristic of K does not divide n . Then, the polynomial $x^n - 1$ splits into linear factors in K if and only if $\mu_n(K) \cong C_n$, or equivalently, there is a primitive n -th root of unity in K .
3. On the other hand, if the characteristic, p , of K divides n , then $x^n - 1$ splits into linear factors if and only if $\mu_n(K) \cong C_m$ for $n = mp^k$ with $p \nmid m$. Equivalently, there is a primitive m -th root of unity in K .

Proof.

This is just the Jacobian criterion.

By separability, we $x^n - 1$ splits into linear factors if and only if it has n distinct roots. Then $\mu_n(K) \cong C_n$, so there is a primitive n -th root of unity. On the other hand having a primitive n -th root of unity means there are n distinct roots.

In the inseparable case, we have $x^n - 1 = (x^m - 1)^{p^k}$ with m, k as defined; then $x^m - 1$ is separable, so that we have a splitting into linear factors if and only if $\mu_n(K) = C_m$, or equivalently, $\mu_m(K) = C_m$. $\square$

Now we can prove the following:

Proposition 3.8.4. Let K be a field, $a \in K$, and let $L := K(b)$ for some b with $b^n = a$.

1. If $x^n - 1$ splits into linear factors in K , then L is a splitting field extension.
2. If K is not of characteristic dividing n , and $a \neq 0$, then L is a separable extension.

3. If (1) is satisfied, then the automorphism group $\text{Aut}_K(L)$ is naturally identified with a subgroup of $\mu_n(K)$. If $x^n - a$ is irreducible, this subgroup is all of $\mu_n(K)$.

Proof. (1) Suppose that $x^n - 1 = (x - 1)(x - \zeta_1) \cdots (x - \zeta_{n-1})$ for some ζ_i . Then $x^n - a = (x - b)(x - b\zeta_1) \cdots (x - b\zeta_{n-1})$.

(2) Thanks to Remark 3.6.6 (which is unnecessary in the case K has characteristic zero, by Corollary 3.6.9), it suffices to show that, if the characteristic of K does not divide n , then the polynomial $x^n - a$ is coprime with its derivative, nx^{n-1} . But for $\text{char } K \neq n$ and $a \neq 0$, working in $K[x]$, we have $(x^n - a, nx^{n-1}) = (x^n - a, x^{n-1}) = (a, x^{n-1}) = (1)$ is the unit ideal. So they are indeed coprime.

(3) Any automorphism must send b to another root of the polynomial $x^n - a$. However, the roots of this polynomial are precisely of the form $b\zeta$ for ζ a root of $x^n - 1$. Now let $\mu_n < K$ denote the subgroup of n -th roots of unity (which need not be isomorphic to C_n). Then we can consider the map $\alpha : \text{Aut}_K(L) \rightarrow \mu_n, \varphi \mapsto \varphi(b)/b$. To see that this is a homomorphism, note that if $\varphi, \psi \in \text{Aut}_K(L)$ with $\varphi(b) = \zeta b, \psi(b) = \xi b$, then $\varphi(\xi b) = \xi \varphi(b) = \xi \zeta b$, as $\xi \in K$. So $\varphi \circ \psi(b) = \zeta \xi b$, and $\alpha(\varphi \circ \psi) = \alpha(\varphi)\alpha(\psi)$. The map is also injective, since a K -linear automorphism of L is determined by the image of b . To see that it is surjective if $x^n - a$ is irreducible, note that in the latter case, $K(b) \cong K[x]/(x^n - b)$, so the map $x \mapsto \zeta x$ is clearly a K -linear automorphism for all $\zeta^n = 1$. $\square$

Summarising these facts, we have:

Corollary 3.8.5. If K does not have characteristic dividing n and contains a primitive n -th root of unity, then the extension $K(b) \supseteq K$ is a Galois extension of degree dividing n (with degree precisely n if $x^n - a$ is irreducible). The Galois group is cyclic.

Actually, we don't need the assumption on the characteristic to talk about the automorphism group; only the part about being separable (hence Galois) requires this.

Now applying Proposition 3.5.6 and Remark 3.5.7 (or, in the separable case, the final statement of Theorem 3.7.1), we obtain (dropping an unnecessary assumption on the characteristic from the lecture):

Corollary 3.8.6. Suppose that $x^n - 1$ splits into linear factors in K . Suppose further that $K \subseteq L \subseteq L(b) \subseteq M$ with $b^n \in K$ for some $n \geq 1$. Then $\text{Aut}_{L(b)}(M) \trianglelefteq \text{Aut}_L(M)$ is a normal subgroup with cyclic quotient.

We say that a group G is solvable if there is a chain of subgroups $\{1\} = G_0 \triangleleft G_1 \triangleleft \cdots \triangleleft G_n = G$ with each G_i normal in G_{i+1} with abelian quotient G_{i+1}/G_i . Now applying the definition of solvable group, we obtain:

Corollary 3.8.7. Let $n \geq 1$, and suppose that $x^n - 1$ splits into linear factors in K . Suppose that there is a chain $K \subseteq K_1 \subseteq \cdots \subseteq K_k = L$ with K_i obtainable from K_{i-1} by adjoining a n_i -th root of an element of K_{i-1} , for $n_i \mid n$. Then $\text{Aut}_K L$ is solvable.

Proof. By the above, in this case the chain $\text{Aut}_{K_i} L$ is a chain of groups normal in the next with cyclic quotient, hence $\text{Aut}_K L$ is solvable. $\square$

Corollary 3.8.8. There cannot be a general formula for the roots of a quintic polynomial in terms of radicals.

The proof relies on the following example:

Example 3.8.9. Let $L = \mathbb{C}(x_1, x_2, x_3, x_4, x_5)$, let S_5 act by permuting variables, and let $K = \mathbb{C}(x_1, x_2, x_3, x_4, x_5)^{S_5}$ be the polynomials invariant under such permutations. We will prove later in the course that an extension $L \supseteq L^G$ is always Galois with group G (provided G acts faithfully on L).

Proof of Corollary 3.8.8. Suppose we have a quintic polynomial in $K[x]$ with nonsolvable Galois (automorphism) group (A_5 or S_5), where K contains all roots of unity and is of characteristic zero. If there were a general formula for the roots in terms of radicals, then the splitting field L of K would be given iteratively adjoining roots. Then this would imply that the Galois group is solvable, a contradiction. But now Example 3.8.9 gives an example of such a quintic polynomial. $\square$

The remainder of this section was not covered in detail in lectures; thus it is optional reading, unless we get back to it in later lectures.

Although we don't need it for Corollary 3.8.8, we can actually drop the hypothesis of $x^n - 1$ splitting into linear factors in Corollary 3.8.7, thanks to the following result:

Lemma 3.8.10. Suppose that $L \supseteq K$ is a splitting field of $x^n - 1$. Then L is a Galois extension of K with abelian Galois group $\text{Aut}_K(L)$.

Proof. If $\text{char } K > 0$, let $p := \text{char } K$, and write $n = p^r m$ for $p \nmid m$. Then $x^n - 1 = (x^m - 1)^{p^r}$, and the splitting field of $x^n - 1$ equals the splitting field of $x^m - 1$, which by Lemma 3.8.3 is separable and of the form $K(\zeta)$ for ζ a primitive m -th root of unity. If $\text{char } K = 0$, then L is automatically a separable extension. In the latter case, set $m := n$, so that, in both cases, $L = K(\zeta)$ is the splitting field of the separable polynomial $x^m - 1$, with ζ a primitive m -th root of unity. So L is a Galois extension. Any automorphism of L over K must send ζ to another primitive m -th root of 1, and this uniquely determines the automorphism of L as $L = K(\zeta)$. Thus, the map $\text{Aut}_K(L) \rightarrow \text{Aut}(\mu_m), \varphi \mapsto \varphi|_{\mu_m}$ is an injective homomorphism, with $\mu_m \subseteq L$ the subgroup of roots of unity, isomorphic to C_m . Note that $C_m \cong \mathbb{Z}/m\mathbb{Z}$ and $\text{Aut}(\mathbb{Z}/m\mathbb{Z}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$, given by ring multiplication, which is abelian. So $\text{Aut}_K(L)$ is abelian. $\square$

Put together, if $L = K(b)$ for $b^n \in K$, then we can take the tower of extensions $K \subseteq K(\zeta) \subseteq L(\zeta)$, for ζ a primitive m -th root of 1 (for $m \mid n$ as in the previous proof) such that $x^n - 1$ splits into linear factors in $K(\zeta)$, and we get that $\text{Aut}_{K(\zeta)}(L(\zeta))$ is solvable, and a normal subgroup of $\text{Aut}_K(L(\zeta))$, with quotient which is abelian. So $\text{Aut}_K(L(\zeta))$ is solvable.

Corollary 3.8.11. There is a quintic polynomial $f \in \mathbb{Q}[x]$ such that no root of f is obtainable by radicals and field operations from $\mathbb{Q}$.

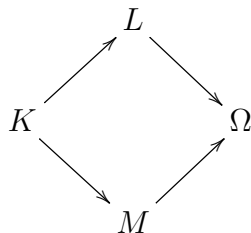
Proof. In Example 8.1.10 later in the course, we will find an explicit polynomial $f \in \mathbb{Q}[x]$ such that the splitting field K of f has Galois group S_5 (or A_5). As this group is not solvable, the splitting field of f cannot be obtainable by extensions using radicals. If there were a root α of f which could be obtained in this way, let $L \supseteq \mathbb{Q}$ be an extension containing α which is obtained from $\mathbb{Q}$ only by extracting radicals. If we further adjoin sufficiently many roots of 1 (taking the splitting field of $x^n - 1$ for n sufficiently large), then by Proposition 3.8.4, we would obtain a splitting field extension M of $\mathbb{Q}$ which contains α . It must therefore contain all roots of f . As this extension required only extracting radicals, and the characteristic is zero, the Galois group $\text{Aut}_{\mathbb{Q}}(M)$ is solvable. But f splits into linear factors in M , so we have a $\mathbb{Q}$ -linear embedding $K \rightarrow M$. We get then a surjective homomorphism $\text{Aut}_M(K) \rightarrow \text{Aut}_{\mathbb{Q}}(K)$. But the image of this is a solvable group, and not the source, which is impossible (any chain of normal subgroups of the source having abelian quotients would map to such a chain for the image). $\square$

Remark 3.8.12. The proof shows more generally: for any field K with polynomial f such that the splitting field L has $\text{Aut}_K(L)$ unsolvable, then no root of f can be obtained from K by extracting radicals along with field operations. We actually don't need to be in the separable situation and require no assumptions on the characteristic of K or on the roots of unity contained therein.

4 Proof of the fundamental theorem of Galois theory

In this section, we will prove Theorem 3.7.1. This relies on the following ingredients:

- Extensions of field embeddings: if $K \rightarrow L, K \rightarrow M$ are field embeddings, with L a finite extension of K , then we can find a common extension Ω of L and M , containing both:



In particular, this will allow us to show that finite normal extensions are splitting field extensions;

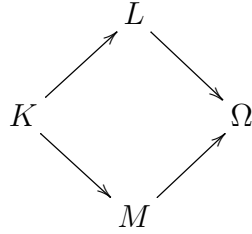
- More on separable extensions: for L a finite extension of K , we define the *separable degree* $[L : K]_s$ to be the number of elements of $\text{Emb}_K(L, \Omega)$ for Ω a normal extension of K for which $\text{Emb}_K(L, \Omega)$ is nonempty. We then show that separable degree satisfies the tower law, and that $[L : K]_s \leq [L : K]$, with equality if and only if L is separable. We also give a better definition of a separable field extension (in terms of field maps rather than elements), and show it is equivalent to the one we gave before;

- Using these theories, we deduce the half $L^{\text{Aut}_K L} = K$ of the Galois correspondence;
- Using separable degree and linear algebra (see Remark 4.5.5 for the idea needed), we show that $\text{Aut}_{L^G}(L) \cong G$, for $G < \text{Aut}(L)$. This gives the other half of the Galois correspondence.

4.1 Extensions of field embeddings

In some expositions of Galois theory, one fixes an algebraic closure of a field K . In this case, every finite extension of K can be viewed as a subfield of the algebraic closure. This is somewhat unappealing from a theoretical point of view, though, because the existence of the algebraic closure requires Zorn's Lemma (equivalently, the axiom of choice), and in practice all one requires is studying a suitable finite extension. The following basic result will allow us to have suitably large finite extensions for any purpose:

Proposition 4.1.1. if $K \rightarrow L, K \rightarrow M$ are field embeddings, with $[L : K] < \infty$, then we can find a common extension Ω of L and M , containing both:

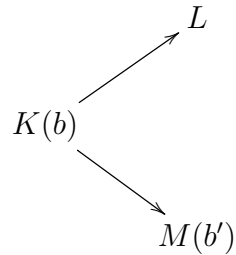


We can take the degree of Ω over M to be at most $[L : K]$, and in particular finite. Alternatively, we can take Ω to be a splitting field over M .

Proof. The proof is by induction on $[L : K]$. The base case is $[L : K] = 1$, implying $L = K$, in which case we can set $\Omega := M$.

Now, if $[L : K] > 1$, let $b \in L \setminus K$ and let $f(x)$ be the minimal polynomial of b over K . We can then form a possibly trivial extension $M(b')$ of M obtained by adjoining a root b' of f (for example we can construct one as $M[x]/(g(x))$ for $g(x)$ any irreducible factor of f). Now, there is an embedding $K(b) \rightarrow M(b')$ sending b to b' : to check this we write $K[x]/(f) \cong K(b)$ and take the map $K[x] \rightarrow M(b')$ sending b to b' . By construction $f(b') = 0$, so we obtain the desired map $K[x]/(f) \rightarrow M(b')$.

We now have a diagram



By the tower law, $[L : K(b)] = [L : K]/[K(b) : K] < [L : K]$, so by the inductive hypothesis, we can find an Ω completing this to a square, and by restriction from $M(b')$ to M and $K(b)$ to K we get the desired diagram.

Moreover, we have $[M(b') : M] \leq [K(b) : K]$. Since, by induction we can take Ω to have $[\Omega : M(b')] \leq [L : K(b)]$, we get from the tower law that $[\Omega : M] \leq [L : K]$, as desired.

For the final statement, we modify the construction by replacing $M(b')$ by a splitting field M' of f over M . By the inductive hypothesis, we have that Ω is a splitting field over M' , which is a splitting field over M , so Ω is a splitting field over M . $\square$

Specialising $M = K$, we immediately obtain the following useful result:

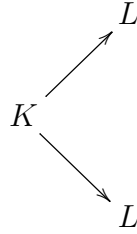
Corollary 4.1.2. For any finite field extension $K \subseteq L$, we can find a further extension $L \subseteq \Omega$ with Ω a splitting field over K .

Observe that, applying this corollary, in the case in Proposition 4.1.1 where both L and M are finite extensions of K , it is possible to take Ω to be a splitting field over K , not merely over M (although the proof of the proposition leads most naturally to the statement made).

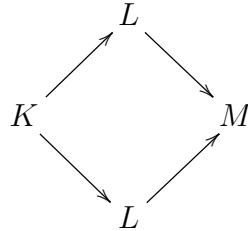
We can also prove the uniqueness of splitting field extensions using this construction:

Corollary 4.1.3. Let L, L' be two splitting fields over K of a polynomial $f \in K[x]$. Then there is a K -linear isomorphism $L \rightarrow L'$.

Proof. Form the diagram



and apply the lemma to obtain a further extension M fitting into



Now, we have the maps $L \rightarrow M \leftarrow L'$, each whose image is a splitting field of f over K . So these images are the same: they are the subfield of M , call it L'' generated by the roots of the polynomial f . We therefore get isomorphisms $L \rightarrow L'' \leftarrow L'$, and inverting the second one gives a composite isomorphism $L \rightarrow L'$. $\square$

4.2 Normality

We can use this to prove that finite normal extensions are splitting field extensions. We prove the following stronger statement:

Proposition 4.2.1. Suppose that $K \subseteq L$ is a **finite** normal extension and $b \in L$. Then the minimal polynomial of b splits into linear factors over L .

Let us first show that this is indeed stronger:

Corollary 4.2.2. If $K \subseteq L$ is a finite normal extension, then L is a splitting field over K .

Proof. If L is finite, then $L = K(\alpha_1, \dots, \alpha_m)$ for some $\alpha_i \in L$. Let f_i be the minimal polynomial of α_i over K . If L is **also** normal, then by Proposition 4.2.1, each f_i splits into linear factors in L . Let $f = \prod_i f_i$. Then f splits into linear factors in L . But also, L is generated by (some of) the roots of f . So L is a splitting field of f over K . $\square$

The proof of Proposition 4.2.1 hinges on the following abstract property of normal extensions:

Lemma 4.2.3. Let $K \subseteq L$ be a normal extension, and suppose that we have a tower of extensions **with $F \subseteq L$ finite**

$$K \subseteq F \subseteq L \subseteq \Omega$$

. Then the following statements hold:

1. The natural injective composition $\text{Emb}_K(F, L) \rightarrow \text{Emb}_K(F, \Omega)$ (post-compose with the inclusion $L \subseteq \Omega$) is also surjective (i.e., bijective);
2. The natural restriction $\text{Emb}_K(L, \Omega) \rightarrow \text{Emb}_K(F, \Omega)$ (pre-compose with the inclusion $F \subseteq L$) is surjective.

What the lemma is saying is as follows: For L a normal extension of K , if there exists an embedding $F \rightarrow L$, then (1) for every $L \subseteq \Omega$ and embedding $F \rightarrow \Omega$, the image lies in L ; (2) moreover, for every embedding $\varphi : F \rightarrow L$, the embedding $F \rightarrow \Omega$ is given by composing φ with some choice of embedding $L \rightarrow \Omega$ (equivalently by normality, the original embedding $L \subseteq \Omega$ composed with an automorphism of L).

Proof. First, we show that (2) implies (1). For each inclusion in the tower $A \rightarrow B$, write $i_{AB} : A \rightarrow B$ as the inclusion. If $\psi : F \rightarrow \Omega$ is an embedding, assuming (2), we can write $\psi = \varphi \circ i_{FL}$ for some $\varphi : L \rightarrow \Omega$. By normality, $\varphi = i_{L\Omega} \circ \theta$ for some automorphism θ of L . Then $\psi = i_{L\Omega} \circ (\theta \circ i_{FL})$, and $\theta \circ i_{FL} \in \text{Emb}_K(F, L)$, so that (1) holds.

It remains to show (2). Fix an embedding $\varphi : F \rightarrow L$, and let $\psi : F \rightarrow \Omega$ be an embedding. We wish to find some $\gamma : L \rightarrow \Omega$ with $\psi = \gamma \circ \varphi$. By Proposition 4.1.1, **since L is**

a finite extension of K , we can find a field $\tilde{\Omega}$ and embeddings $K, \Omega \rightarrow \tilde{\Omega}$ form a commutative diagram of finite extensions,

$$\begin{array}{ccc} L & \xrightarrow{\tilde{\psi}} & \tilde{\Omega} \\ \varphi \uparrow & & \uparrow \tilde{\varphi} \\ F & \xrightarrow{\psi} & \Omega. \end{array}$$

Now, there are two embeddings $L \rightarrow \tilde{\Omega}$: one by $\tilde{\psi}$ in the diagram, the other by $\tilde{\varphi} \circ i_{L\Omega}$ (using the original embedding $L \subseteq \Omega$). By normality, these two differ by an automorphism $\gamma : L \rightarrow L$, so that $\tilde{\psi} = \tilde{\varphi} \circ i_{L\Omega} \circ \gamma$. Let $\tilde{\gamma} = i_{L\Omega \circ \gamma}$. We get a diagram:

$$\begin{array}{ccc} L & \xrightarrow{\tilde{\psi}} & \tilde{\Omega} \\ \varphi \uparrow & \searrow \tilde{\gamma} & \uparrow \tilde{\varphi} \\ F & \xrightarrow{\psi} & \Omega. \end{array}$$

where the square and the upper triangle commute. We claim the lower triangle also commutes. We have that both maps $F \rightarrow \Omega$ (namely, ψ and $\tilde{\gamma} \circ \varphi$) are the same after composing with $\tilde{\varphi}$. But the latter is an embedding, so we get $\psi = \tilde{\gamma} \circ \varphi$. In particular, $\tilde{\gamma}$ gives the desired embedding. $\square$

Proof of Proposition 4.2.1. Let $K \subseteq L$ be a finite normal extension and $b \in L$ have minimal polynomial f . Form a splitting field Ω of f over L . For $K(b) \subseteq L$ the subfield generated by b over K , we have the tower

$$K \subseteq K(b) \subseteq L \subseteq \Omega.$$

Now, as L is normal, every embedding $K(b) \rightarrow \Omega$ lands in L . Let $b' \in \Omega$ be a root of f . We can then form the map $K(b) \cong K[x]/(f) \rightarrow \Omega, b \mapsto b'$. By the preceding, we find that $b' \in L$. Thus, all roots of f are in L , and f splits into linear factors over L . $\square$

Lemma 4.2.3 is actually much more powerful than Proposition 4.2.1, and also allows us to prove the surjectivity of $\text{Aut}_K(M)/\text{Aut}_L(M) \rightarrow \text{Aut}_K(L)$ for normal extensions (Remark 3.5.7:

Corollary 4.2.4. Take a tower $K \subseteq L \subseteq M$ of extensions with $K \subseteq M$ and $K \subseteq L$ normal, and $L \subseteq M$ finite. Then the natural injective homomorphism $\text{Aut}_K(M)/\text{Aut}_L(M) \rightarrow \text{Aut}_K(L)$ is surjective.

We thus get $\text{Aut}_K(M)/\text{Aut}_L(M) \cong \text{Aut}_K(L)$ as desired (and verifying the final assertion of Theorem 3.7.1).

Proof. Apply Lemma 4.2.3.(2) to the tower $K \subseteq L \subseteq M \subseteq M$ (setting F to L and L and Ω to M). We obtain that the restriction map $\text{Aut}_K(M) \rightarrow \text{Emb}_K(L, M)$ is surjective (as we already saw, it is well-defined by normality of L). Now for every $\varphi \in \text{Aut}_K L$, composing with the inclusion $i : L \subseteq M$, we get an embedding $L \rightarrow M$. By the first sentence, this extends to an automorphism of M which restricts on L to $i \circ \varphi$, as desired. $\square$

Example 4.2.5. Let $K := \mathbb{Q}(\zeta)$ for ζ a primitive n -th root of unity. Let $L = K(b)$ be obtained by adjoining a root b of $x^n - 2$ (irreducible by Eisenstein's criterion). Then we showed that L is normal and $\text{Aut}_K L \cong C_n$, given by transformations satisfying $b \mapsto \zeta^a b$ for all $0 \leq a < n$. We can take the subfield $K(b^m)$ for any $m \mid n$. The same reasoning then shows that $\text{Aut}_K K(b^m) \cong C_{n/m}$. On the other hand, $\text{Aut}_{K(b^m)} L$ consists of those automorphisms of the form $b \mapsto \zeta^{an/m} b$ for some $a \geq 0$. So we get $\text{Aut}_{K(b^m)} L = C_m \leq C_n$. The corollary then correctly asserts that $C_n/C_m \cong C_{n/m}$.

4.3 Separable extensions and degree

The motivating goal for this section is to prove the equivalence of various definitions of separability (notably, Remark 3.6.7 and Definition 3.6.5). To do this, we will make more precise the connection between separable polynomials, i.e., distinct roots, and the number of embeddings. The resulting notion of *separable degree* will turn out to be much more important than the proposition itself, similarly to how Lemma 4.2.3 is more important than the equivalence of various definitions of normality (Proposition 4.2.1).

Roughly speaking the idea is as follows: if $f \in K[x]$ is an irreducible polynomial, and L is a splitting polynomial of f over K , then the number of embeddings $K[x]/(f) \rightarrow L$ is equal to the number of distinct roots of f : $x + (f)$ can map to any of these. So if the roots are all distinct, then the number of embeddings equals the degree of f , hence of the field extension $[K[x]/(f) : K]$.

Definition 4.3.1. Let $K \subseteq L$ be a finite extension, and let $\Omega \supseteq L$ be a further **finite** extension such that $\Omega \supseteq K$ is normal. Then the separable degree $[L : K]_s$ is defined as the number of K -linear embeddings $L \rightarrow \Omega$.

Note that an Ω exists thanks to Corollary 4.1.2.

Lemma 4.3.2. This definition does not depend on the choice of Ω .

Proof. Suppose that Ω_1, Ω_2 are two finite extensions of L which are normal over K . By Proposition 4.1.1, we can form a field Ω admitting L -linear embeddings of both Ω_1 and Ω_2 . By Lemma 4.2.3, we obtain by composition with these embeddings bijections $\text{Emb}_K(L, \Omega_1) \rightarrow \text{Emb}_K(L, \Omega) \leftarrow \text{Emb}_K(L, \Omega_2)$. $\square$

This also makes a nice connection to normal extensions:

Corollary 4.3.3. A finite extension $K \subseteq L$ has $|\text{Aut}_K(L)| \leq [L : K]_s$, with equality if the extension is normal. If the extension is not normal and $[L : K]_s$ is finite, then $|\text{Aut}_K(L)| < [L : K]_s$.

Proof. If L is normal, then we can take $\Omega = L$ in the definition of $[L : K]_s$, obtaining the equality. On the other hand, if L is not normal, let Ω be a field extension such that there are two K -linear embeddings $\varphi_1, \varphi_2 : L \rightarrow \Omega$ with different images. By composing automorphisms of L with the inclusion into Ω , we obtain an injection $\text{Aut}_K(L) \rightarrow \text{Emb}_K(L, \Omega)$, but the assumption shows us that this is not surjective. We obtain in particular the inequality. When $[L : K]_s$ is finite, these sets are finite, so $[L : K]_s > |\text{Aut}_K(L)|$. $\square$

Example 4.3.4. Let's take $K := \mathbb{Q}(\sqrt[n]{2})$ for some $n \geq 3$. The minimal polynomial of $\sqrt[n]{2}$ is $x^n - 2$, since this polynomial is irreducible by Eisenstein's criterion. So $[K : \mathbb{Q}] = n$. But $K \supseteq \mathbb{Q}$ is not a normal extension, since $K \subseteq \mathbb{R}$ and hence has no other roots of $x^n - 2$. Thus $\text{Aut}_{\mathbb{Q}} K = \{1\}$ is trivial. But for every root b of $x^n - 2$, there is an embedding $K \rightarrow \mathbb{Q}(\sqrt[n]{2}, \zeta)$ with ζ a primitive n -th root of unity, sending $\sqrt[n]{2}$ to b . So $[K : \mathbb{Q}]_s = n = [K : \mathbb{Q}]$. Note that K is a separable extension because the roots of $x^n - 2$ are distinct (or because $\mathbb{Q}$ is perfect).

Theorem 4.3.5. The following are equivalent for a finite extension $K \subseteq L$:

1. The minimal polynomial of every element of L is separable (Definition 3.6.5);
2. L is generated over K by elements $\alpha_1, \dots, \alpha_m$ such that the minimal polynomial of α_i over $K(\alpha_1, \dots, \alpha_{i-1})$ is separable;
3. $[L : K]_s = [L : K]$;
4. For every intermediate extension $K \subseteq K_1 \subsetneq K_2 \subseteq L$, there exists a field M and two distinct K_1 -linear embeddings $K_2 \rightarrow M$ (Remark 3.6.7).

Before we prove Theorem 4.3.5, we develop the theory of separable degree a bit:

Proposition 4.3.6. 1. For a tower $K \subseteq L \subseteq M$, we have $[M : K]_s = [M : L]_s [L : K]_s$ (the tower law);

2. We have $[L : K]_s \leq [L : K]$.

Proof. (1) We prove this by induction on $[L : K]$. For the base case, if this is one, then $L = K$ and there is nothing to prove. If $[L : K] > 1$, let $b \in L$ be an element with minimal polynomial of degree $n > 1$. Then we can consider the tower $K(b) \subseteq L \subseteq M$. By the tower law for ordinary degree, $[L : K(b)] < [L : K]$. So by the inductive hypothesis, $[M : K(b)]_s = [M : L]_s [L : K(b)]_s$. It remains to show $[L : K]_s = [L : K(b)]_s [K(b) : K]_s = n [L : K(b)]_s$.

Let L' be a normal extension of K admitting an embedding $L \rightarrow L'$. By definition, we have to show that $|\text{Emb}_K(L, L')| = |\text{Emb}_{K(b)}(L, L')| |\text{Emb}_K(K(b), L')|$. Consider the restriction map

$$\pi : \text{Emb}_K(L, L') \rightarrow \text{Emb}_K(K(b), L').$$

The fibre of π over an embedding $K(b) \rightarrow L'$ consists of precisely the $K(b)$ -linear embeddings $L \rightarrow L'$, $\text{Emb}_{K(b)}(L, L')$, for this embedding. Since all fibres have the same size, we get the desired formula.

(2) Using the tower laws, we can reduce this statement to the case of an extension $L = K(b)$ generated by a single element. Let the minimal polynomial of b be $f \in K[x]$ and let $n = \deg f$. We have to show that $[K(b) : K]_s \leq n$. But every embedding is uniquely determined by the image of b , which must be a root of f , and there are at most n choices. $\square$

Proof of Theorem 4.3.5. By Proposition 4.3.6, the equality (3) ($[L : K]_s = [L : K]$) holds if and only if for each intermediate extension $K \subseteq K_1 \subseteq K_1(b) \subseteq L$, we have that the minimal polynomial of b over K_1 is separable. Now, the minimal polynomial of b over K_1 is a factor of

the minimal polynomial over K , so this is in turn equivalent to (1). Given what we just said, it is clear that (1) implies (2). Given (2), applying Proposition 4.3.6.(2) and the ordinary tower law, we get (3). So (1),(2),(3) are all equivalent.

Now, suppose that (3) holds. Consider a chain $K \subseteq K_1 \subsetneq K_2 \subseteq L$. By the tower law, $[K_2 : K_1]_s > 1$, so that (4) holds.

To finish, we show that (4) implies (2). By the tower law, it suffices to assume that $L = K(b)$ for some element b . Now assume that (2) does not hold. Let f be the minimal polynomial of b . Then f is inseparable, so by Proposition 3.6.10, for $\text{char } K = p$, we have $f(x) = g(x^p)$ for some polynomial g . We then have that b^p is a root of g . So we can consider the extension $K \subseteq K(b^p) \subseteq K(b)$. Since $\deg g < \deg f$ and b^p is a root of g , we have $[K(b^p) : K] < [K(b) : K]$. We will show that for every M into which $K(b)$ embeds, the restriction map $\text{Emb}_K(K(b), M) \rightarrow \text{Emb}_K(K(b^p), M)$ is bijective, hence (4) does not hold, establishing the contrapositive we need. Indeed, since we are in characteristic p , p -th roots are unique when they exist (if c is a root of $x^p - a$ then $x^p - a = (x - c)^p$ by the Freshner's Dream). So, given an embedding $K(b^p) \rightarrow K$, if the image of b^p has a p -th root, it is unique, establishing a unique embedding $K(b) \rightarrow K$. $\square$

Combined with Corollary 4.3.3, we have the following numerical criteria for Galois extensions:

Corollary 4.3.7. For a finite extension $K \subseteq L$, we have $|\text{Aut}_K(L)| \leq [L : K]_s \leq [L : K]$. The first equality holds if and only if the extension is normal, the second if and only if the extension is separable, and both hold if and only if the extension is Galois.

4.4 The easy half of the fundamental theorem: $L^{\text{Aut}_K L} = K$

Theorem 4.4.1. For $K \subseteq L$ a Galois extension, we have $L^{\text{Aut}_K L} = K$.

In the context of a Galois extension $F \subseteq L$ and an intermediate field K , we obtain that the composition $K \mapsto \text{Aut}_K L \mapsto L^{\text{Aut}_K L}$ is the identity.

Proof of Theorem 4.5.1. The inclusion $\text{Aut}_{L^{\text{Aut}_K L}} L \leq \text{Aut}_K L$ is an equality, since every automorphism in $\text{Aut}_K L$ is $L^{\text{Aut}_K L}$ -linear. Since L is normal, $[L : L^{\text{Aut}_K L}]_s = [L : K]_s = |\text{Aut}_K L|$. By the tower law for separable degree, $[L^{\text{Aut}_K L} : K]_s = 1$. But $L^{\text{Aut}_K L} \supseteq K$ is separable, as it is intermediate to a separable extension. So $L^{\text{Aut}_K L} = L$. $\square$

Remark 4.4.2. A slightly different argument is: By Lemma 4.2.3, every embedding $L^{\text{Aut}_K L} \rightarrow L$ is the composition of the standard inclusion with a K -linear automorphism of L . As all of the latter are the identity on $L^{\text{Aut}_K L}$, we get that $[L^{\text{Aut}_K L} : K]_s = 1$. But $L^{\text{Aut}_K L} \supseteq K$ is separable, as it is intermediate in a separable one. So $[L^{\text{Aut}_K L} : K] = 1$, so $K = L^{\text{Aut}_K L}$.

4.5 The other half: $\text{Aut}_{L^G}(L) = G$

In this section we prove the following theorem:

Theorem 4.5.1. Let $G < \text{Aut}(L)$ be finite. Then the extension $L \supseteq L^G$ is Galois, and the natural map $G \rightarrow \text{Aut}_{L^G}(L)$ is an isomorphism.

Specialising this to the case that $K \subseteq L$ is a Galois extension and replacing G by $H \leq \text{Aut}_K L$, we obtain that the composition $H \mapsto L^H \mapsto \text{Aut}_{L^H} L$ is the identity. Before we prove the theorem, we record the useful consequence:

Corollary 4.5.2. For any finite extension $K \subseteq L$, we have $|\text{Aut}_K L| \mid [L : K]$.

Proof. By Theorem 4.5.1, we have that $[L : L^{\text{Aut}_K L}] = |\text{Aut}_K L|$. The result then follows from the tower law. $\square$

Proof of Theorem 4.5.1. First, note that every $g \in G < \text{Aut}(L)$ is the identity on L^G , so that actually $G < \text{Aut}_{L^G}(L)$, and the inclusion map $G \rightarrow \text{Aut}_{L^G}(L)$ is an injective homomorphism. We then have $|G| \leq |\text{Aut}_{L^G}(L)| \leq [L : L^G]_s \leq [L : L^G]$. So to prove the assertions, in view of Corollary 4.3.7, we only need to prove the inequality $[L : L^G] \leq |G|$.

This we will prove using linear algebra over L together with the Galois group. To prove this we do the following clever argument (for an explanation of why it works, see Remark 4.5.5). Let $G = \{g_1, \dots, g_n\}$, for $g_1 = 1$. We wish to show that any $n + 1$ elements $b_1, \dots, b_{n+1} \in L$ are linearly dependent over K . To this end, we will turn this into a linear problem over L . Define the embedding

$$\varphi : L \rightarrow L^n, \quad b \mapsto (g_1 b, g_2 b, \dots, g_n b).$$

For dimension reasons, $\varphi(b_1), \dots, \varphi(b_{n+1})$ are linearly dependent over L . Suppose that $m \leq n + 1$ is minimal such that we have a linear dependence amongst m of these. Without loss of generality, we may reorder the b_i so that the relation is

$$a_1 \varphi(b_1) + \dots + a_m \varphi(b_m) = 0, a_i \in L.$$

Rescaling, we may assume that $a_1 = 1$. We can rewrite this as

$$g(b_1) + a_2 g(b_2) + \dots + a_m g(b_m) = 0, \forall g \in G. \quad (4.5.3)$$

Now, applying $h \in G$, we get also

$$h g(b_1) + h(a_1) h g(b_2) + \dots + h(a_m) h g(b_m) = 0, \forall g \in G.$$

This is the same thing as saying

$$g(b_1) + h(a_2) g(b_2) + \dots + h(a_m) g(b_m) = 0, \forall g \in G. \quad (4.5.4)$$

Subtracting (4.5.4) from (4.5.3), we get a new linear relation, which involves fewer of the b_i . By our assumption, this is zero, so we obtain that

$$h(a_i) = a_i, \forall i,$$

which just means $a_i \in K$ for all K . So we have a linear dependence over K , not merely over L . Now setting $g = 1$ we get the desired dependence

$$b_1 + a_2 b_2 + \dots + a_m b_m = 0. \quad \square$$

Remark 4.5.5. What the proof is actually doing is using the Galois group to compare linear algebra over K with linear algebra over L . In general, given a Galois extension L over K with Galois group $G = \text{Aut}_K L$, and given any K -vector space V , we can form the L -vector space of V by extension of scalars, formally $V_L := V \otimes_K L$. This L -vector space is naturally equipped with additional structure: the action of G on L by K -linear automorphisms extends to an action on $V \otimes_K L$ by $g(v \otimes b) := v \otimes g(b)$. This action is not L -linear, but rather satisfies the compatibility condition: for $w \in V_L$, $a \in L$, and $g \in G$, $g(aw) = g(a)g(w)$. We call such an action of G on an L -vector space V_L a *descent datum*. The maps $V \mapsto V \otimes_K L$ and $W \mapsto W^G$ give mutually inverse equivalences (of categories!):

$$\{K\text{-vector spaces}\} \rightarrow \{L\text{-vector spaces with a descent datum}\}.$$

The above proof uses this idea (though we need to prove the extension is Galois): we wish to find a nonzero vector in the kernel of the map $T : K^{n+1} \rightarrow L$, $(a_1, \dots, a_{n+1}) \mapsto a_1 b_1 + \dots + a_{n+1} b_{n+1}$. We replace this map by the L -linear map $\tilde{T} : L^{n+1} \rightarrow L^n$, $(a_1, \dots, a_{n+1}) \mapsto a_1 \varphi(b_1) + \dots + a_{n+1} \varphi(b_{n+1})$. This obviously has a kernel, which is G -invariant, and we show the G -invariant subspace is nonzero, which is then a kernel of the original map T . (In fact, one can equip L^n with a nonstandard G -action such that $(L^n)^G = L$ (rather than K^n), and taking G -invariant subspaces literally restricts $\tilde{T}$ to T : for $g \in G$, let $\sigma_g \in S_n$ be the permutation such that $g^{-1} g_j = g_{\sigma_g(j)}$. Then we can consider the action $g \cdot (b_1, \dots, b_n) = (g b_{\sigma_g(1)}, \dots, g b_{\sigma_g(n)})$. This still has the property $g \cdot (c(b_1, \dots, b_n)) = g(c)g \cdot (b_1, \dots, b_n)$, so it is a descent datum satisfying $(L^n)^G = L$.)

Note that the two halves, Theorems 4.4.1, 4.5.1, together with Proposition 3.5.6 and Corollary 4.2.4, imply all of Theorem 3.7.1.

4.6 The inverse Galois problem

Once we have the fundamental theorem, the following question becomes of key interest: Given a finite group G , how can we realise G as the Galois group of a Galois extension $K \subseteq L$? We have the following basic existence result:

Proposition 4.6.1. Let K be an arbitrary field and let $L := K(x_1, \dots, x_n)$ be the field of rational functions in the x_i . Then $S_n \leq \text{Aut}_K L$. For every $G < S_n$, we have that L is a Galois extension of L^G of Galois group G . Moreover, L is the splitting field over L^G of a polynomial of degree n .

Proof. Note that $S_n \subseteq \text{Aut}_K L$. Now, thanks to Theorem 4.5.1, we have that $L \supseteq L^G$ is a Galois extension with Galois group G . Finally, we have that $(x - x_1) \cdots (x - x_n)$ has coefficients which are invariant under S_n so in $L^{S_n} \subseteq L^G$, and the splitting field is clearly L . $\square$

Applying this to the case of $n = 5$ we immediately get:

Corollary 4.6.2. For any characteristic, there is a Galois extension $L \supseteq K$ of that characteristic with L the splitting field of a quintic polynomial over K , such that the Galois group is S_5 , or A_5 , and hence unsolvable.

5 Interpretation as permutations of roots

5.1 The symmetric group

We have one more important connection to make that we mentioned at the beginning: Given a splitting field L of a polynomial f over K , we asserted that $\text{Aut}_K L$ is connected to permutations of the roots of f . Indeed, since the automorphisms are K -linear, they preserve f , and hence the roots of f (or equivalently, factors of f over L , all linear). Since L is generated by the roots, this action uniquely determines the automorphism of L . So we get an injective homomorphism

$$\Phi : \text{Aut}_K L \hookrightarrow \text{Perm}(\{\text{roots of } f\}). \quad (5.1.1)$$

In the case that f is separable, the latter group is isomorphic to $S_{\deg f}$ (all roots are distinct), otherwise it could be a smaller permutation group.

Thus, one profitable way to compute automorphism groups (sometimes called Galois groups, at least when $L \supseteq K$ is Galois) is to find the image of Φ , which is isomorphic to $\text{Aut}_K L$. A useful piece of information in this process is the following:

Proposition 5.1.2. Let L be a finite normal extension of K . Suppose that $f \in K[x]$ splits into linear factors in L . Then f is irreducible if and only if $\text{Aut}_K L$ acts transitively on the roots: that is, for any two roots α, β of f , there exists an element $\varphi \in \text{Aut}_K L$ such that $\varphi(\alpha) = \beta$.

Proof. First if $\text{Aut}_K L$ acts transitively on the roots, then f must be irreducible, since $\text{Aut}_K L$ has to preserve the roots of any factor $g \mid f$ with $g \in K[x]$, as it preserves the coefficients. We prove the converse by extension of embeddings. Take the extension $K(\alpha)$. Let $i : K(\alpha) \rightarrow L$ be the inclusion. Note that $K(\alpha) \cong K[x]/(f(x))$ because f is irreducible, hence the minimal polynomial of α up to nonzero scaling). Then we have another embedding $\phi : K(\alpha) \cong K[x]/(f(x)) \rightarrow L$ sending α to β . By Lemma 4.2.3, we have that $\phi = \psi \circ i$, for some automorphism $\psi \in \text{Aut}_K L$. This automorphism ψ has the desired property that $\psi(\alpha) = \beta$. $\square$

Another way to interpret this is that *the set of roots of the minimal polynomial f of α in any Galois extension containing α is the orbit of α under the Galois group of the extension.*

This has a useful numerical consequence: If you remember from group theory, given a group G acting transitively on a set X , for any $x \in X$ we can form the stabiliser $G_x := \{g \in G \mid g(x) = x\}$, and counting gives the “orbit-stabiliser formula”, $|G| = |X||G_x|$. So in particular, $|X| \mid |G|$. In the case at hand, this yields:

Corollary 5.1.3. If L is the splitting field of a separable irreducible polynomial $f \in K[x]$, then $\deg f$ divides $|\text{Aut}_K L|$.

Of course, we already knew this though since L is a Galois extension of K , so $|\text{Aut}_K L| = [L : K]$, and the latter is a multiple of $\deg f$ by the tower law: $[K(\alpha) : K] = \deg f$ for α a root of f .

5.2 The discriminant and the alternating group

It turns out that, at least over the rational numbers, “most” polynomials of degree n yield the full symmetric group as the Galois group.

One of the main question in Galois theory is to determine which subgroup of the symmetric group one obtains for a given polynomial. A first natural question is: *is the subgroup contained in the alternating group?* We will answer that here.

Given a monic polynomial $f(x) \in K[x]$, let L be a splitting field of the polynomial and write it there as $f = \prod (x - \alpha_i)$ for α_i the roots. Then we form the *discriminant*,

$$\Delta_f := \prod_{i < j} (\alpha_i - \alpha_j)^2. \quad (5.2.1)$$

The discriminant is nonzero if and only if the roots are distinct, i.e., f is separable.

Remark 5.2.2. (*Non-examinable:*) This hints at the fact that we can also define the discriminant using the resultant of f and f' , see Section 8.3 or Commutative Algebra for more details on the resultant—this implies that $\Delta_f \in K$, but we give another proof using Galois theory below. The resultant gives more though: if $f(x) \in R[x]$, then in fact $\Delta_f \in R$, for any integral domain R .

Assume that f is separable, equivalently that Δ_f is nonzero. In this case L is separable, and hence Galois. We note that Δ_f is clearly invariant under the Galois group, so $\Delta_f \in K$. Also, Δ_f is a square in L . Let $\sqrt{\Delta_f}$ denote either of the square roots, i.e.,

$$\sqrt{\Delta_f} := \pm \prod_{i < j} (\alpha_i - \alpha_j) \in L,$$

for either choice of sign (anyway, without the $\pm$ the RHS depends on the ordering of the roots). We then have a subfield $K(\sqrt{\Delta_f}) \subseteq L$. Note that if we act on $\sqrt{\Delta_f}$ by a permutation, we have

$$\sigma(\sqrt{\Delta_f}) = \text{sign}(\sigma)(\sqrt{\Delta_f}). \quad (5.2.3)$$

Therefore, for $G := \text{Aut}_K L \leq S_n$, an element $g \in G$ fixes $\sqrt{\Delta_f}$ if and only if $g \in A_n$. Therefore we obtain:

Theorem 5.2.4. Suppose that $\Delta_f \neq 0$. Then K is Galois. The Galois group $G \leq S_n$ is in A_n if and only if Δ_f is a square.

Proof. We already saw the first two statements. We just saw that $\sqrt{\Delta_f} \in L$ is fixed by the entire group G if and only if $G \leq A_n$. But $L^G = K$. $\square$

Example 5.2.5. We compute the discriminant of the cubic $f(x) = x^3 + 3px + 2q$: as derived in Problems Sheet 2, the roots are (Cardano’s formula):

$$\omega^i \sqrt[3]{-q + \sqrt{q^2 + p^3}} + \omega^{-i} \sqrt[3]{-q - \sqrt{q^2 + p^3}}, \quad i \in \{0, 1, -1\},$$

Here, ω is a cube root of unity, and the cube roots [and square roots] have to be chosen so that the product of the two summands is $-p$ (instead of $-\omega p$ or $-\omega^2 p$). We now compute the discriminant. Let $A := \sqrt[3]{-q + \sqrt{q^2 + p^3}}$ and $B := \sqrt[3]{-q - \sqrt{q^2 + p^3}}$. We have:

$$\begin{aligned}\Delta_f &= (A + B - \omega A - \omega^{-1} B)^2 (A + B - \omega^{-1} A - \omega B)^2 (\omega A + \omega^{-1} B - \omega^{-1} A - \omega B)^2 \\ &= (1 - \omega)^6 (A - \omega^{-1} B)^2 (-\omega^{-1} A + B)^2 (\omega A - \omega B)^2 = (-3\omega)^3 ((A - \omega^{-1} B)(A - \omega B)(A - B))^2 \\ &= -27(A^3 - B^3) = -27(2\sqrt{q^2 + p^3})^2 = -108(q^2 + p^3). \quad (5.2.6)\end{aligned}$$

The coefficient and sign is *absolutely* important (as you saw in problems class)!

We remark that the above is all valid for an arbitrary field since Cardano's formula is (as you can verify). But for $\text{char } K = 2, 3$, the discriminant is zero, i.e., there are always two equal roots (which is easy to see from the formula for f). If you want to take a general monic cubic $x^3 + bx^2 + cx + d$ you will get:

$$b^2 c^2 - 4c^3 - 4b^3 d - 27d^2 + 18bcd,$$

which does not have to be zero in characteristics two and three.

Example 5.2.7. Consider the cubic polynomial $x^3 + 3px$. The discriminant is $-108p^3$, which is a square if and only if $-3p$ is a square. Indeed, we factor the polynomial as $x(x - \sqrt{3p})(x + \sqrt{3p})$, so this checks out!

Next consider the case $x^3 + 2q$. Then the discriminant is $-108q^2$, which is a square if and only if -3 is a square. Indeed, the roots are $\sqrt[3]{2q}, \omega\sqrt[3]{2q}, \omega^2\sqrt[3]{2q}$ for ω a primitive cube root of unity (assume the characteristic is not three, otherwise the three roots are the same and the discriminant is zero). That means that we always get ω in the splitting field. This satisfies a quadratic polynomial, $x^2 + x + 1$, with roots $\frac{-1 \pm \sqrt{-3}}{2}$. So the quadratic extension is indeed nontrivial if and only if -3 is not a square, as desired.

Example 5.2.8. For a quadratic $x^2 + bx + c = (x - \alpha)(x - \beta)$, the discriminant is $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = b^2 - 4c$, as you know from high school. There it is usually for $ax^2 + bx + c$ and the discriminant is written as $b^2 - 4ac = a^2(\alpha - \beta)^2$ for α, β the roots. Indeed, for a non-monic polynomial $f = a_n x^n + \dots$ of degree n , by convention one defines

$$\Delta_f := a_n^{n-2} \prod_{i < j} (\alpha_i - \alpha_j)^2 = a_n^{n-2} \Delta_{f/a_n}.$$

The discriminant here is a square if and only if the field extension is trivial, i.e., the roots are already in K . It is moreover zero if and only if the roots are the same. In particular, in the case of $K = \mathbb{R}$, we see that the discriminant is a square if and only if it is nonnegative. Thus we get a nontrivial extension ($\mathbb{C}$) if and only if it is negative.

Remark 5.2.9. If you look carefully at the above examples of discriminant, you will see that in characteristic two, it is *always* a square. This is true in general, and here is why: suppose that it were nonzero. Then the roots are distinct, so L is a Galois extension. But

if the discriminant were not a square, then $K(\sqrt{\Delta_f})$ would be a nontrivial extension of K . As it is over characteristic two and given by adjoining a square root, it is inseparable (square roots are unique in characteristic two). But all subextensions of a Galois extension are separable, contradiction. An alternative proof is that, the same argument that shows that Δ_f is invariant under the Galois group shows, in characteristic two, that $\sqrt{\Delta_f}$ is also invariant, so that $\sqrt{\Delta_f}$ actually has to be in the ground field if the extension is Galois.

6 Biquadratic extensions

In this section we address completely the simplest type of Galois extensions which go beyond cyclic ones. Amazingly this can be done using only two square roots (so in particular, these extensions are contained in the extension field of constructible lengths using a compass and straightedge we referred to earlier in the notes).

In the words of Alessio Corti, this example is a *pièce de resistance* which every beginner in Galois theory needs to master completely.

6.1 The main statement

We will study the field extension

$$L = K(\sqrt{a \pm \sqrt{b}}), a, b \in K, \quad (6.1.1)$$

We will assume that b is not a square in K , otherwise this extension is not very interesting. (On the other hand, the extension is interesting for $a = 0$.) Then, L is the field obtained by adjoining square roots of $a \pm \sqrt{b}$, where $\sqrt{b}$ is any square root of b (it does not matter which one). Assuming b is not a square, this is the smallest normal extension which contains the extension $K(\sqrt{a + \sqrt{b}})$, as we have a K -linear automorphism of $K(\sqrt{b})$ given by sending $\sqrt{b}$ to $-\sqrt{b}$, and this will extend to a K -linear isomorphism $K(\sqrt{a + \sqrt{b}}) \rightarrow K(\sqrt{a - \sqrt{b}})$; to be normal then a field containing $\sqrt{a + \sqrt{b}}$ also must contain $\sqrt{a - \sqrt{b}}$ (or more simply, the two have the same minimal polynomial).

We begin with the following observations.

- (i) The extension L is the splitting field of the polynomial, identifiable as a quadratic polynomial in x^2 ,

$$f(x) = (x^2 - a)^2 - b = x^4 - 2ax^2 + c, \quad c := a^2 - b. \quad (6.1.2)$$

- (ii) This extension is separable if the characteristic of K is not equal to two. That is because $f' = 4x^3 - 4ax$, and if the characteristic is not two, its roots are $0, \pm\sqrt{a}$, which are not roots of f .

(iii) Consider the following identity:

$$(\sqrt{a + \sqrt{b}} + \sqrt{a - \sqrt{b}})^2 = 2a + 2\sqrt{a^2 - b} = 2a + 2\sqrt{c}. \quad (6.1.3)$$

So the field L also contains the similar expressions $\sqrt{2a \pm \sqrt{4c}}$. Note that if we iterate this construction we get back $2\sqrt{a \pm \sqrt{b}}$, so this actually shows $L = K(\sqrt{2a \pm \sqrt{4c}})$. In other words, L is also the splitting field of the “companion polynomial”,

$$g(y) = y^4 - 4ay^2 + 4b, \quad (6.1.4)$$

and the companion polynomial of g is $16f(x/2)$.

We see above that it is very convenient to assume *the characteristic of K is not two*. This is because of the constant issues of factors of two appearing (which I do not believe you can remove).

Let D_4 denote the dihedral group of order 8, i.e., the rotational and reflectional symmetries of the square. (This is sometimes denoted D_8 in the literature, but I am trying to use the geometric notation.)

Theorem 6.1.5. Let K be a field of characteristic not equal to two. Let $a, b \in K$ with b not a square. Consider the Galois extension (6.1.1). Let G be the Galois group of $L \supseteq K$ and set $c := a^2 - b$. Then there is an injection $G \hookrightarrow D_4$, and we have

- (i) If bc, c are not squares in K , then $[L : K] = 8$ so that $G = D_4$;
- (ii) If bc is a square in K (so c is not a square in K since b is not), then $[L : K] = 4$ and $G = C_4$;
- (iii) If c is a square in K (so bc is not), then *either*
 - (1) Neither $2(a + \sqrt{c})$ or $2(a - \sqrt{c})$ are squares in K . Then $[L : K] = 4$ and $G \cong C_2 \times C_2$; or
 - (2) One of $2(a \pm \sqrt{c})$ is a square (but not both). Then $L = K(\sqrt{b})$ and $G \cong C_2$.

6.2 Key elements of L

Here we replace our somewhat imprecise notation of square roots by defining elements in L . This simplifies the observations we made before and sets up the notation we need for the intermediate fields that appear.

Let $\beta \in L$ be such that $\beta^2 = b$ (we are just being more precise now than using $\sqrt{b}$). Next, let $\alpha, \alpha' \in L$ be such that $\alpha^2 = a + \beta$ and $(\alpha')^2 = a - \beta$. Then $L = K(\alpha, \alpha')$ by definition, and the roots of f are $\pm\alpha, \pm\alpha' \in L$.

We will use the following quantities:

1. $\gamma := \alpha\alpha'$. Note that $\gamma^2 = (a + \beta)(a - \beta) = a^2 - b = c$. So $\gamma \in L$ and $K(\gamma) = K(\sqrt{c})$ is a quadratic extension of K , contained in L ;

2. $\delta := \alpha + \alpha'$ and $\delta' := \alpha - \alpha'$. Then as we already observed,

$$\delta^2 = \alpha^2 + (\alpha')^2 + 2\gamma = 2(a + \gamma); \quad (\delta')^2 = \alpha^2 + (\alpha')^2 - 2\gamma = 2(a - \gamma).$$

In particular, $L = K(\delta, \delta')$. Note that $\pm\delta, \pm\delta'$ are the roots of the companion polynomial $g(y)$.

3. Finally, $\delta\delta' = 2\beta$:

$$\delta\delta' = (\alpha + \alpha')(\alpha - \alpha') = \alpha^2 - (\alpha')^2 = a + \beta - (a - \beta) = 2\beta.$$

Exercise 6.2.1. Write formulas for γ, α, α' in terms of δ, δ' .

6.3 The action of G on the roots of f

We now get to the heart of the computation of G and hence the structure of the extension L . As in previous examples, what we really want to do is to construct an embedding of G into D_4 , so that we can then proceed to identify the image. We remark that, since L is the splitting field of a quartic, we could have proceeded simply by first embedding G into S_4 then identifying the image as D_4 .

In any case, to begin with, we wish to find what the automorphisms of L might possibly do to the set of roots $\alpha, -\alpha, \alpha', -\alpha'$.

Proposition 6.3.1. Every automorphism of L preserves the following square (i.e., acts by rotation or reflection on it):

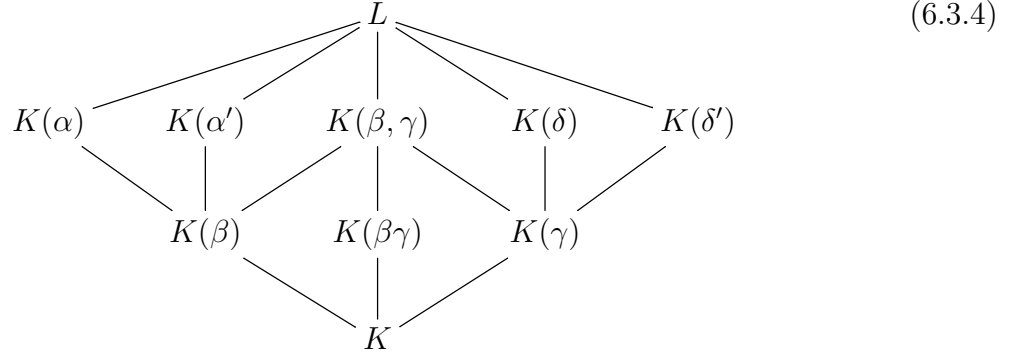
$$\begin{array}{ccc} & \alpha' & \\ & / \quad \backslash & \\ -\alpha & & \alpha \\ & \backslash \quad / & \\ & -\alpha' & \end{array} \tag{6.3.2}$$

In particular, the injection $G \hookrightarrow S_4$ has image inside D_4 .

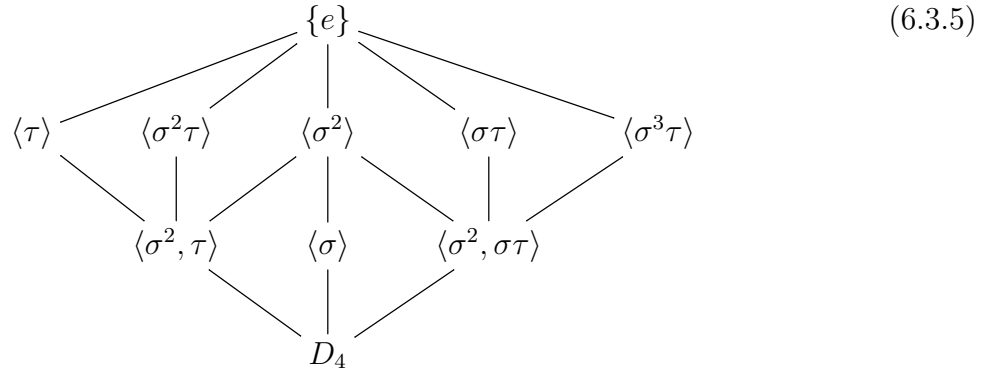
Proof. It is enough to show that every automorphism sends opposite vertices to opposite vertices. This is clear because opposite vertices are negatives. $\square$

Corollary 6.3.3. The intermediate fields of $L \supseteq K$ are a subset of the ones in the following

diagram of inclusions (“Hasse diagram”). We have every field here if and only if $G = D_4$.



Proof. We write out the “Hasse diagram” of subgroups of D_4 , generated by the counterclockwise rotation σ by $\pi/2$ and the reflection τ about the horizontal axis (the defining relations are $\sigma^4 = 1 = \tau^2, \tau\sigma\tau^{-1} = \sigma^{-1}$):



Now, it is an easy check from the definitions that the fixed fields for the subgroups which are in the image of the Galois group contain the fields in the diagram above. It is a little more work to check that they are the entire fixed subfields. One way to see this though is as follows: All of the adjacent fields have degree at most two, since the larger field (on the top) is obtained from the smaller one (on the bottom) by adjoining a square root. Now, if any of the fixed fields were larger than indicated above, then take such a field K' as high up as possible. This field cannot be L as L cannot get any larger. Take any field $K'' \supseteq K'$ which is one row up from K' . As $[K'' : K'] \leq 2$, the only way to make K' larger while fixing K'' is to let $K' = K''$. In other words, if any of the fixed fields were larger than indicated, then we would end up with the same fixed field for two adjacent subgroups of D_4 . But we can check that for each of these adjacent subgroups, there is an element fixed by the smaller subgroup but not by the larger one (an explicit expression in the elements above). Namely, any listed generator for the larger field will have this property. $\square$

6.4 Proof of Theorem 6.1.5

We obtain from the preceding the following more precise version of Theorem 6.1.5:

Corollary 6.4.1. The possible images of $\text{Aut}_K L$ in D_4 are:

- (i) All of D_4 , in the case bc, c are not squares in K ;
- (ii) $\langle \sigma \rangle \cong C_4$, in the case that bc is a square (so c is not);
- (iii) $\langle \sigma^2, \sigma\tau \rangle \cong C_2 \times C_2$, in the case that c is a square and $2(a \pm \sqrt{c})$ are not;
- (iv) $\langle \sigma\tau \rangle \cong C_2$, in the case that c and $2(a + \sqrt{c})$ are squares;
- (v) $\langle \sigma^3\tau \rangle \cong C_2$, in the case that c and $2(a - \sqrt{c})$ are squares.

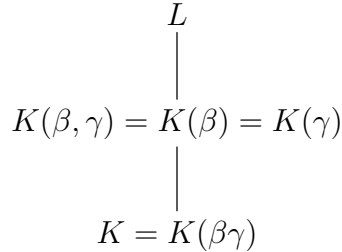
Proof. Given Corollary 6.3.3, the subset of the Hasse diagram that we get is easy to determine: they are just the fixed fields L^H for $H \leq G$. In turn, G has the property that $K = L^G$. So if G is not all of D_4 we get that certain elements appearing in the diagram are already in K . The possibilities are restricted because of our assumption that b is not a square, hence $\beta \notin K$. So the only possible listed generators which could be in K are $\beta\gamma, \gamma, \delta$, and δ' . These correspond to the Galois groups $C_4, C_2 \times C_2, C_2$, and C_2 respectively, and the case for these to be in K are precisely when bc is a square, c is a square but not $2(a \pm \sqrt{c})$, and when c is a square and one of $2(a \pm \sqrt{c})$ is a square, respectively. $\square$

6.5 The explicit Hasse diagrams

We can write out the Hasse diagrams in each of the cases (i)–(v). Recall here and below that $\beta = \sqrt{b}, \gamma = \sqrt{c}, \delta = \sqrt{2(a + \sqrt{c})}, \delta' = \sqrt{2(a - \sqrt{c})}$.

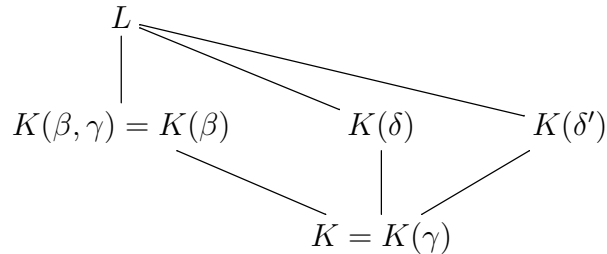
Case (i) is just (6.3.4).

Case (ii) is the subset with $K(\beta\gamma) = K$ on bottom:



In this situation, we claim that $L = K(\delta) = K(\delta') = K(\alpha) = K(\alpha')$. To see this we note that $K(\delta) = K(\delta, \gamma) = K(\delta, \beta) = K(\delta, \delta') = L$, and similarly $K(\delta') = L$; then we have $K(\alpha) = K(\alpha, \beta) = K(\alpha, \gamma) = K(\alpha, \alpha') = L$ and similarly $K(\alpha') = L$.

Case (iii) is the subset with $K(\gamma)$ on the bottom:



In this case we claim that $L = K(\alpha) = K(\alpha')$. Indeed, since $\gamma \in K$ and $\alpha\alpha' = \gamma$, we have $K(\alpha) = K(\alpha') = K(\alpha, \alpha') = L$.

Case (iv) is the subset with $K(\delta)$ on the bottom:

$$\begin{array}{c} L = K(\beta) \\ \searrow \\ K = K(\delta) \end{array}$$

Case (v) is the subset with $K(\delta')$ on the bottom:

$$\begin{array}{c} L = K(\beta) \\ \searrow \\ K = K(\delta') \end{array}$$

6.6 Examples

Example 6.6.1. If K is the splitting field of $x^4 - 2$ over $\mathbb{Q}$, then $a = 0, c = -2, b = 2, bc = -4$, so $G = D_4$ and we get the full Hasse diagram.

Example 6.6.2. If K is the splitting field of $x^4 - 4x^2 + 2$ over $\mathbb{Q}$, then $a = 2, c = 2, b = 2, bc = 4$, so we get $G = C_4$ and $K = \mathbb{Q}(\sqrt{2 + \sqrt{2}})$.

Example 6.6.3. If K is the splitting field of the cyclotomic polynomial $x^4 - x^2 + 1$ whose roots are primitive twelfth roots of unity (called $\Phi_{12}(x)$) of $\mathbb{Q}$, then $G = C_2 \times C_2$. (Here $a = 1/2, c = 1, b = -3/4, bc = -3/4, 2(a \pm \sqrt{c}) = 1 \pm 2 = \{-1, 3\}$.) This matches what we will prove later, that the Galois group is identified with $(\mathbb{Z}/12\mathbb{Z})^\times$.

Example 6.6.4. If K is the splitting field of the polynomial $x^4 - 10x^2 + 1$ over $\mathbb{Q}$ (here $a = 5, c = 1, b = 24, bc = 24, 2(a \pm \sqrt{c}) = \{8, 12\}$), then $G = C_2 \times C_2$. In this case, $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which explains the identity, popular in secondary school (at least where I did it),

$$\sqrt{5 + 2\sqrt{6}} = \sqrt{2} + \sqrt{3}.$$

At least, Galois theory implies that the LHS must be of the form $A\sqrt{2} + B\sqrt{3}$ for some $A, B \in \mathbb{Q}$.

7 The fundamental theorem of algebra

We will now give a simple proof of the fundamental theorem of algebra.

7.1 Preliminaries from analysis

As we discussed, any proof of the fundamental theorem of algebra must involve some analysis, because in spite of its name and the statement which is algebraic, the truth of the theorem involves the analytic definition and nature of $\mathbb{C}$.

Theorem 7.1.1. If $f \in \mathbb{R}[x]$ has odd degree, then there exists a real root of f .

Proof. (This is nonexaminable.) Write it as $a_n x^n + \cdots + a_0$ for $n = \deg f$ odd. We have that $\lim_{x \rightarrow \infty} f(x) = \infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$, as can be seen because $\lim_{x \rightarrow \pm\infty} f(x)/x^n = a_n \neq 0$. Now there exists values $x_0 \ll 0, x_1 \gg 0$ with $f(x_0) < 0 < f(x_1)$. Then the intermediate value theorem shows that for some $x \in [x_0, x_1]$, $f(x) = 0$. $\square$

Proposition 7.1.2. Every quadratic polynomial $f \in \mathbb{C}[x]$ has a root.

Proof. (This is nonexaminable.) Thanks to the quadratic formula (or completing the square), it suffices to show that every complex number $z \in \mathbb{C}$ has a square root. Taking the polar decomposition $z = re^{i\theta}$ (for $r \geq 0$ and $e^{i\theta} := \cos \theta + i \sin \theta$), a square root has the form $\sqrt{r}e^{i\theta/2}$, reducing the question to the fact that every nonnegative number has a real square root (this can be seen directly from the definition or axioms of the real numbers). $\square$

7.2 Preliminaries from algebra

Theorem 7.2.1 (First Sylow theorem). Let G be a finite group, and suppose $|G| = p^m r$ for p a prime, $m \geq 1$, and $p \nmid r$. Then there exists a subgroup $H < G$ with $|H| = p^m$.

Such a subgroup H is called a *Sylow p -subgroup* of G . This is just a subgroup of order the maximum possible power of p for every prime divisor p of $|G|$.

Proof. (This is non-examinable.) Let $X := \{S \subseteq G \mid |S| = p^m\}$ be the set of subsets of size p^m . We have an action $G \times X \rightarrow X$ given by left multiplication. Note that $|X| = \binom{p^m r}{p^m}$, which is *not* a multiple of p , because the maximum power of p dividing n with $1 \leq n \leq p^m$ is the same as the maximum power dividing $n + p^m(r - 1)$. (Or, one can use the lemma $\binom{pa}{pb} \equiv \binom{a}{b} \pmod{p}$, coming from $(x+1)^{pa} \equiv (x^p+1)^a \pmod{p}$, to get $\binom{p^m r}{p^m} \equiv r \pmod{p}$.)

So there is an element $S \in X$ such that the orbit $\mathcal{O} = G \cdot S \subseteq X$ has size not a multiple of p . By the orbit-stabiliser theorem, the stabiliser $H \leq G$ of S has size a multiple of p^m . But $|H| = |HS| \leq |X|$ since $HS \subseteq S$, thus $|H| = p^m$, giving the desired subgroup. $\square$

We also require a basic result on p -groups, defined as groups of order a power of p (such as Sylow p -subgroups).

Proposition 7.2.2. Let G be a finite group of order p^m for some prime p and $m \geq 1$. Then there exists a subgroup $H < G$ of index p .

Proof. (This is non-examinable.) We first prove that $Z(G)$ is nontrivial. Consider the action of G on itself by conjugation, $G \times G \rightarrow G, (g, h) \mapsto ghg^{-1} \in G$. The orbits here are the conjugacy classes. By the orbit-stabiliser theorem, all conjugacy classes have order a factor of $|G|$. In the case at hand that means they all have order a power of p . Now, the sum of the sizes of the conjugacy classes equals $|G| = p^m$, which is a multiple of p . This means that the number of conjugacy classes of size 1 is a multiple of p . That is, the size of the centre is a multiple of p . In particular the centre is nontrivial.

Now we prove the assertion by induction on $|G|$. The base case is $|G| = p$, where the result is clear. For the inductive step, we have that $G/Z(G)$ is also a p -group of smaller size. By the inductive hypothesis, $G/Z(G)$ has a subgroup of index p . Its preimage in G has index p as well (by the third isomorphism theorem, or just by counting). $\square$

7.3 The proof of the fundamental theorem of algebra

Theorem 7.3.1 (Fundamental theorem of algebra). If $f \in \mathbb{C}[x]$ is a polynomial of positive degree, then f has a complex root.

Proof. It is equivalent to show that every finite extension $\mathbb{C} \subseteq L$ has $L = \mathbb{C}$. Let L be such a finite extension. Then we also have $\mathbb{R} \subseteq L$. By Corollary 4.1.2, L is a subfield of a splitting field of $\mathbb{R}$. Let us replace L by such a splitting field. Since we are in characteristic zero, L is in fact Galois over $\mathbb{R}$.

By Theorem 7.1.1, the minimal polynomial of every $b \in L$ over $\mathbb{R}$ must have even degree. Now, let $H < \text{Aut}_{\mathbb{R}} L$ be a Sylow 2-subgroup. Then by Theorem 4.5.1, $[L : L^H] = |H|$, and by the tower law, $[L^H : \mathbb{R}]$ is odd. By the fact in the first sentence, this can only happen if $[L^H : \mathbb{R}] = 1$, i.e., $L^H = \mathbb{R}$. So $[L : \mathbb{R}]$, and hence also $[L : \mathbb{C}]$, are powers of two.

Write $[L : \mathbb{C}] = 2^m$ for some $m \geq 0$. We wish to show $m = 0$. For a contradiction, suppose that $m \geq 1$. Let $G := \text{Aut}_{\mathbb{C}} L$. By Proposition 7.2.2, there is a subgroup $H \leq G$ of index 2. Then, $[L^H : \mathbb{C}] = 2$. We obtain a nontrivial quadratic extension of $\mathbb{C}$. But this is impossible by Proposition 7.1.2. $\square$

8 Frobenius lifting

8.1 Preliminaries on finite fields

This was contained in CW1, #5, but it is worth going over:

Let K be a finite field. Then the unique ring homomorphism $\varphi : \mathbb{Z} \rightarrow K$ (determined from $1 \mapsto 1$) is not injective, so that the characteristic of K is not zero. Therefore it must be a prime $p \geq 2$. In particular, $K \supseteq \mathbb{F}_p := \mathbb{Z}/p\mathbb{Z}$, which is the image of φ . Now, as K is finite, it is a finite-dimensional vector space over $\mathbb{F}_p$, i.e., has finite degree $d := [K : \mathbb{F}_p]$, and thus $|K| = p^d$ for some $d \geq 1$. Conversely, we have the following:

Proposition 8.1.1. There is a unique finite field of order p^d up to isomorphism, for every prime p and $d \geq 1$.

This classifies all finite fields up to isomorphism.

Proof. Suppose that $|K| = p^d$. Then $|K^\times| = p^d - 1$, but $K^\times$ is cyclic, as it is a finite subgroup of the multiplicative group of a field. So $x^{p^d-1} = 1$ for all $x \in K^\times$. Thus every element of K is a root of the polynomial $x^{p^d} - x$. But this polynomial can have at most $p^d = |K|$ roots. So $x^{p^d} - x$ is separable with roots equal precisely to K . In particular, K is a splitting field of $x^{p^d} - x$ over $\mathbb{F}_p$. Conversely, we know that splitting fields exist (Proposition 3.5.2) and are unique (Corollary 4.1.3). $\square$

Definition 8.1.2. Let $q := p^d$. Then we define $\mathbb{F}_q$ as any finite field of order q (unique up to isomorphism).

To study the Galois groups, we need the Frobenius map:

Definition 8.1.3. Let K be a field of characteristic p . Then the Frobenius map $\text{Fr}_p : K \rightarrow K$ is defined by $\text{Fr}_p(x) := x^p$. Similarly, define $\text{Fr}_{p^d} := \text{Fr}_p^d$, the map sending x to x^{p^d} .

Note that Fr_p is a field homomorphism, hence injective. But it need not be surjective:

Example 8.1.4. $\text{Fr}_p(\mathbb{F}_p(x)) = \mathbb{F}_p(x^p) \subsetneq \mathbb{F}_p(x)$.

Proposition 8.1.5. For K a field with $|K| = p^d$, we have that $\text{Fr}_p(K) = K$. So $\text{Fr}_p \in \text{Aut}(K)$. Moreover, the order of Fr_p is d .

Proof. The first statement follows because any injective map from a finite set to itself must be surjective. Then, $\text{Fr}_{p^d} = \text{Fr}_p^d$ is the identity on K since $x^q = x$ for all $x \in K$ as we saw above. Finally, for $d' \mid d$, then $\text{Fr}_{p^{d'}}$ cannot be the identity on K , since $\text{Fr}_{p^{d'}}(x) = x$ implies that x is a root of $x^{p^{d'}} - x$, but there can be at most $p^{d'} < p^d$ roots. $\square$

Proposition 8.1.6. Let $K \subseteq L$ be an extension of finite fields. Then this extension is Galois with cyclic Galois group C_d for $d = [L : K]$, generated by the map Fr_q for $q = |K|$. Moreover, K is the unique subfield of L of size $|K| = q$, given as the fixed field $K = L^{\text{Fr}_q}$.

Proof. We already saw that Fr_q is the identity on K and is an automorphism of L . Hence $\text{Fr}_q \in \text{Aut}_K L$. But we also saw that Fr_q has order d . So $C_d \leq \text{Aut}_K L$. Since $|\text{Aut}_K L| \leq [L : K] = d$, we get $C_d \cong \text{Aut}_K L$ and $|\text{Aut}_K L| = [L : K]$, so L is a Galois extension of K with Galois group C_d . It then follows that $K = L^{\text{Aut}_K L} = L^{\text{Fr}_q}$. $\square$

This is very useful, as we will see a bit later, for the following reason:

Corollary 8.1.7. Let $|K| = q$ and let $f \in K[x]$ be an irreducible polynomial. Let L be a finite normal extension in which f splits into linear factors. Then the Frobenius morphism Fr_q is an automorphism of L which induces a cyclic permutation of the set of roots of f .

Proof. We already saw that Fr_q is an automorphism of L with fixed field K , which generates $\text{Aut}_K L$. The latter group must act transitively on the set of roots of f thanks to Proposition 5.1.2. Since it is generated by Fr_q , the only possibility is that Fr_q induces a cyclic permutation of the roots. $\square$

In particular, if K is the splitting field of a (separable) polynomial $f = f_1 \cdots f_k \in \mathbb{F}_p[x]$ with distinct monic irreducible factors f_i of degrees d_i , then Fr_p induces a permutation of the roots of f with cycle type $(d_1)(d_2) \cdots (d_k)$.

In this section, we will prove:

Theorem 8.1.8. Let $f \in \mathbb{Z}[x]$ be a polynomial which reduces to a separable polynomial $f_p \in \mathbb{F}_p[x]$ modulo p , and let $K \supseteq \mathbb{Q}$, $F \subseteq \mathbb{F}_p$ be splitting fields of f and f_p . Then there exists a bijection $\psi : Z \rightarrow Z_p$ between the roots Z of f and Z_p of f_p and a subgroup $H \leq \text{Aut}_{\mathbb{Q}} K$ such that we have an isomorphism $\iota : H \rightarrow \text{Aut}_{\mathbb{F}_p} F$ with the property that $\iota(h) \circ \psi = \psi \circ h$.

Put together with Corollary 8.1.7 and the following discussion, we get the result:

Corollary 8.1.9. If $f \in \mathbb{Z}[x]$ is such that the reduction $f_p \in \mathbb{F}_p[x]$ is separable and decomposes into irreducible polynomials of degree $n_1, \dots, n_k$, then the image of the injection $\text{Aut}_{\mathbb{Q}} K \hookrightarrow S_{\deg f}$ includes an element of cycle type $(n_1) \cdots (n_k)$.

Example 8.1.10. We show that the splitting field K over $\mathbb{Q}$ of the polynomial $x^5 - x - 1$ has Galois group S_5 . Modulo $p = 2$, the polynomial factors as $x^5 + x + 1 = (x^2 + x + 1)(x^3 + x^2 + 1) \in \mathbb{F}_2[x]$, with both factors irreducible. So the Galois group $\text{Aut}_{\mathbb{Q}} K$, by the above, has an element of cycle type $(2)(3)$. Next, modulo $p = 3$, the polynomial is irreducible, as we can check because it does not have a root and is not divisible by any irreducible quadratic modulo 3 (there are only three to check). So applying the above we also get a five-cycle in $\text{Aut}_{\mathbb{Q}} K$. Now, we can see that any subgroup of S_5 containing both a five-cycle and an element of cycle type $(2)(3)$ is the entire group (first of all, the latter is odd, so the subgroup is not contained in A_5 , the only subgroup of index two; it is generated by elements of order 5 and 6 and is not abelian so its order is a multiple of 30 but not 30 itself; so it must have order 120 and be the entire group. Or you can prove this in various other ways.) As a result, we obtain the polynomial claimed in Corollary 3.8.11 and get that the roots of f are not obtainable from $\mathbb{Q}$ using only field operations and radicals.

8.2 Preliminaries on Dedekind's independence theorem

In order to proceed, we will need the following classical result about automorphisms of fields. Actually it is much more general, and concerns *characters* valued in fields. The most general setting of this is for characters of semigroups:

Definition 8.2.1. A semigroup S is a set together with an associative binary operation $S \times S \rightarrow S$. Homomorphisms of semigroups are just multiplicative maps. If S has a unit element, then S is called a monoid; monoid homomorphisms are semigroup homomorphisms sending the unit to the unit.

Example 8.2.2. Any group is a monoid. Any ring is a monoid under multiplication (and an abelian group under addition); fields are commutative monoids under multiplication.

Given a semigroup S and a field K viewed as a semigroup under multiplication, we can consider semigroup homomorphisms $S \rightarrow K$. These are called *characters (of S valued in K)*.

Example 8.2.3. Suppose that $S = G$ is a group. Then a character $G \rightarrow K$ lands in $K^\times = \text{GL}_1(K)$. Such characters are also called *one-dimensional matrix representations* in group representation theory.

Example 8.2.4. Any field automorphism $\varphi : K \rightarrow K$ is also a character of K valued in K . This can also be interpreted as a character $K^\times \rightarrow K$ with domain $K^\times$, (or with target $K^\times$).

Example 8.2.5. For any semigroup, we can consider two distinguished characters: one that sends everything to zero (called the *zero character*), and another that sends everything to one (called the *trivial character*). Note that for a monoid, a character χ cannot be zero, since $\chi(1) = 1$.

Theorem 8.2.6 (Linear independence of characters, or Dedekind's independence theorem). Let S be a semigroup and K a field. Any set $\chi_1, \dots, \chi_n : S \rightarrow K$ of distinct nonzero characters is a linearly independent subset of the K -vector space of functions $S \rightarrow K$.

It's clear that to have linear independence, we need the assumption about characters being nonzero. Note again that this assumption is automatic if we use a monoid (or a group!) instead of a semigroup.

Proof. This is a similar proof by induction and contradiction to Theorem 4.5.1. The statement is clear if $n = 1$ (because we don't allow χ_i to be identically zero). Assume the statement for $n \leq N$. Now assume for a contradiction that there is a linear dependence,

$$\sum_{i=1}^{N+1} \lambda_i \chi_i = 0 \text{ for } \lambda_i \in K. \quad (8.2.7)$$

We can assume by induction that all $\lambda_i \neq 0$. By the inductive hypothesis, $\chi_1, \dots, \chi_N$ are linearly independent, so the dependence in (8.2.7) is unique up to nonzero scaling.

Next find $c \in M$ such that $\chi_1(c) \neq \chi_2(c)$, and consider the new relation

$$\sum_i \lambda_i \chi_i(c) \chi_i = 0, \quad (8.2.8)$$

which holds because $\sum_i \lambda_i \chi_i(cd) = \sum_i \lambda_i \chi_i(c) \chi_i(d)$. As before, this is the unique linear dependence up to nonzero scaling. We get that $\lambda_i \chi_i(c) / \lambda_i = \chi_i(c)$ does not depend on i , contradicting our construction of c . $\square$

Example 8.2.9. We can use this theorem to recover the inequality $|\text{Aut}_K L| \leq [L : K]$ for finite extensions (part of Corollary 4.3.7). As we saw in Example 8.2.4, an automorphism $\sigma \in \text{Aut}_K L$ is in particular a character of L valued in L . So they are linearly independent over L . Since the characters are K -linear, though, they are also elements of the vector space of K -linear transformations, $\text{Hom}_{K\text{-lin}}(L, L)$. This is an L -vector space under the multiplication $(a\varphi)(b) = a(\varphi(b))$. Its dimension over L is $[L : K]$ (since $L \cong K^{[L:K]}$ as a K vector space, so $\text{Hom}_{K\text{-lin}}(L, L) \cong \text{Hom}_K(K^{[L:K]}, L) \cong L^{[L:K]}$, now as L -vector spaces). By linear independence, we get $|\text{Aut}_K L| \leq [L : K]$.

8.3 Preliminaries on algebraic integers

Definition 8.3.1. An algebraic integer is the root of a monic polynomial $f \in \mathbb{Z}[x]$.

Proposition 8.3.2. An element α is an algebraic integer if and only if its minimal polynomial over $\mathbb{Q}$ is in $\mathbb{Z}[x]$.

The proposition easily follows from the following useful lemma:

Lemma 8.3.3. Suppose that $f, g \in \mathbb{Q}[x]$ are monic and $g \mid f$. If $f \in \mathbb{Z}[x]$, then also $g \in \mathbb{Z}[x]$.

Remark 8.3.4. Note that the statement and proof also are valid with $\mathbb{Z} \subseteq \mathbb{Q}$ replaced by any UFD inside its field of fractions.

Proof of Lemma 8.3.3. Since g is monic, by clearing denominators we can find $m \geq 1$ with $mg \in \mathbb{Z}[x]$ a polynomial having coprime coefficients. Write $f = (mg)h$ for $h \in \mathbb{Q}[x]$. Since f, mg are integral with coprime coefficients, by Gauss's Lemma, so is h (see Section 2.2). Now comparing leading coefficients, it follows that $m \mid \pm 1$ and hence $m = 1$. $\square$

Proposition 8.3.5. The sum and product of two algebraic integers is still an algebraic integer.

Proof. Given two algebraic integers α, β , write their minimal polynomials over $\mathbb{Q}$ as $f = \prod_i (x - \alpha_i), g = \prod_j (x - \beta_j)$. By the above, these are in $\mathbb{Z}[x]$. Consider the polynomial

$$h := \prod_{i,j} (x - \alpha_i - \beta_j).$$

This is monic. Observe that its coefficients are rational, because for $K = \mathbb{Q}(\alpha_i, \beta_j)$, a normal extension of $\mathbb{Q}$, we have that h is fixed under $\text{Aut}_{\mathbb{Q}} K$.

We claim that its coefficients are integers. To see this, we make use of the resultant. This is covered in Commutative Algebra, to which we refer the interested student (or to internet search), but we recall briefly the definition and properties. Suppose that $f, g \in R[x]$ for some integral domain R . Then the resultant $\text{Res}(f, g) \in R$ is a certain determinant, whose entries are coefficients of f and g , such that $\text{Res}(f, g) = 0$ if and only if f and g share a common factor. Namely, f and g share a common factor if and only if the equations $fa + gb = 0$ have a nonzero solution for $\deg(a) < \deg(g)$ and $\deg(b) < \deg(f)$. Writing $a = a_0 + a_1x + \cdots + a_{\deg(g)-1}x^{\deg(g)-1}$ and $b = b_0 + b_1x + \cdots + b_{\deg(f)-1}x^{\deg(f)-1}$, we can regard the a_i and b_i as variables, and $fa + gb = 0$ as a system of $\deg(f) + \deg(g)$ linear equations in these $\deg(f) + \deg(g)$ variables. So the determinant of the matrix of coefficients has the desired property.

Now let us work over the ring $R = \mathbb{Z}[y]$, and take $H := \text{Res}(f(x), g(y - x)) \in R = \mathbb{Z}[y]$. We don't expect H to be zero: it's basically impossible for $f(x)$ and $g(y - x)$ to have a common root in $\mathbb{Q}[y]$ since the roots of $f(x)$ are constant in y , unlike those of $g(y - x)$. But, for any homomorphism $\varphi : R = \mathbb{Z}[y] \rightarrow K$, with $\varphi(y) = \gamma \in K$, we get that $\text{Res}(f(x), g(\gamma - x)) = \varphi(H) = H(\gamma)$, because taking determinant commutes with homomorphisms of the underlying

ring. Now $\varphi(H) \in K$ is zero if and only if $f(x)$ and $g(\gamma - x)$ have a common root. This is definitely possible!

Taking K to be a splitting field of fg over $\mathbb{Q}$, we get that $H(\gamma) = 0$ if and only if $\alpha_i = \gamma - \beta_j$ for some i, j , i.e., if and only if $\gamma = \alpha_i + \beta_j$, i.e., γ is a root of h . Thus, $h(y)$ and $H(y)$ have the same roots. It follows from the definition of h that $h \mid H$. By Lemma 8.3.3, since h, H are monic, $H \in \mathbb{Z}[x]$, and $h \in \mathbb{Q}[x]$, we must have $h \in \mathbb{Z}[x]$, as desired. (Actually we can do better: by inspecting the determinant which defines H , we see that $\deg(H) \leq \deg(f) \deg(g) = \deg(h)$, so that $H = h$.)

Similarly, we have that $\prod_{i,j} (x - \alpha_i \beta_j)$ divides (and in fact equals) the resultant over $R = \mathbb{Q}[y]$ of $f(x)$ and $y^{\deg(g)} g(y/x)$, which has integral coefficients. $\square$

Remark 8.3.6. *Omit from lecture, non-examinable:* Here is another proof which is perhaps more standard (perhaps, more conceptually appealing, but a bit less practically appealing): Claim: the following are equivalent:

- (i) α is an algebraic integer;
- (ii) $\mathbb{Z}[\alpha]$ is finitely generated $\mathbb{Z}$ -module;
- (iii) There is a finitely generated free $\mathbb{Z}$ -module V with an action of $\mathbb{Z}[\alpha]$ (i.e., a $\mathbb{Z}[\alpha]$ -module);
- (iv) α is an eigenvalue of a square matrix with integral coefficients.

It is immediate that (i) implies (ii), since α^n is an integral combination of $1, \alpha, \dots, \alpha^{n-1}$ for some n . Also, (ii) implies (iii) by taking the module to be $\mathbb{Z}[\alpha]$, which is free over $\mathbb{Z}$ because it is a submodule of $\mathbb{Q}[\alpha]$, which is torsion free, and the fundamental theorem of $\mathbb{Z}$ -modules says that every torsion-free finitely-generated $\mathbb{Z}$ -module is free. Also, to see that (iv) implies (i), note that the characteristic polynomial g of A is monic, and by the Cayley–Hamilton theorem, $g(A) = 0$, so the same is true for every eigenvalue.

So we only need to show (iii) implies (iv). To see this, suppose that V is a $\mathbb{Z}[\alpha]$ -module with a $\mathbb{Z}$ -basis $v_1, \dots, v_n$. Let A be the matrix for multiplication by α in terms of this basis. Then A has integral coefficients. For f the minimal polynomial of α , we have $f(A) = 0$, since A acts by multiplication by α . So the minimal polynomial of A (i.e., the monic generator of the ideal of polynomials g such that $g(A) = 0$) is a factor of f , but since f is irreducible, f is the minimal polynomial of A . Next, the roots of the minimal polynomial of A are the eigenvalues of A (by linear algebra). So that means that α is indeed an eigenvalue of A .

Now we can use this to prove the proposition. We use the Kronecker product to construct, from square matrices A, B with eigenvalues α, β , new square matrices with eigenvalues $\alpha + \beta$ and $\alpha\beta$. Recall that the Kronecker product of matrices A, B with entries $(a_{ij})_{i,j=1}^m$ and $(b_{kl})_{k,\ell=1}^n$ is the matrix with entries $(a_{ij}b_{kl})_{i,j=1; k,\ell=1}^{m;n}$, denoted $A \otimes B$, where now we use two indices for the rows and for the columns. To be pedantic, this matrix has $(i + mk, j + m\ell)$ -th entry $a_{ij}b_{kl}$ for $1 \leq i, j \leq m$ and $1 \leq k, \ell \leq n$. We can verify explicitly that, if $Av = \alpha v$ and $Bw = \beta w$, then $(A \otimes B)(v \otimes w) = \alpha\beta(v \otimes w)$.

Now, if A, B have integral entries, it follows that $A \otimes B$ also does. We saw it has eigenvalue $\alpha\beta$, showing this is an algebraic integer. Similarly, $(A \otimes I_n) + (I_m \otimes B)$ has integral entries and has eigenvalue $\alpha + \beta$ (for the same eigenvector $v \otimes w$ as before). So $\alpha + \beta$ is an algebraic integer.

Really the Kronecker product is a matrix version of the tensor product, which is a nicer way to write things without using bases. In these terms, we have $(A \otimes B)(v \otimes w) = Av \otimes Bw$ by definition!

Finally, if you don't like Kronecker products or tensor products, there is an explicit way to construct them in the situation at hand: if f, g are the minimal polynomial of α, β of degrees m, n , then multiplication by x on $\mathbb{Z}[\alpha]$ has minimal polynomial f , so has α as an eigenvalue and there is some eigenvector $v \in \mathbb{C}[x]/(f)$ with $xv = \alpha v$; similarly we have $w \in \mathbb{C}[y]/(g)$ with $yw = \beta w$. Then in $\mathbb{C}[x, y]/(f, g)$ we have $(x + y)(vw) = (\alpha + \beta)(vw)$ and $(xy)(vw) = (\alpha\beta)(vw)$. But $x + y$ and xy define integral matrices in the basis of monomials $x^i y^j, i < m, j < n$. So again $\alpha + \beta, \alpha\beta$ are the eigenvalues of integral square matrices.

Remark 8.3.7. *Omit from lecture, non-examinable:* Instead of using resultants in the original proof, we could also have argued as follows that the coefficients of h are integral: the coefficients of h are symmetric polynomials in α and β . We can also think of them as symmetric polynomials in the α_i whose coefficients are symmetric polynomials in the β_j . Then the Fundamental Theorem of Symmetric Polynomials says that every symmetric polynomial with integral coefficients is a polynomial in the elementary symmetric polynomials—the coefficients of f and g (up to sign). So this shows that the coefficients of h are polynomials in the coefficients of f and g with integral coefficients, hence are integral.

8.4 Frobenius lifting and computing Galois groups modulo p

Theorem 8.4.1. Let $f(x) \in \mathbb{Z}[x]$ be degree n and monic, and let $K \supseteq \mathbb{Q}$ be a splitting field of f . Finally let $p \geq 2$ be a prime number such that the reduction $f_p \in \mathbb{F}_p[x]$ of $f \bmod p$ has n distinct roots, with $F \supseteq \mathbb{F}_p$ a splitting field of f_p . Let $\lambda_1, \dots, \lambda_n \in K$ be the roots of f . Define $R := \mathbb{Z}[\lambda_1, \dots, \lambda_n] \subset K$. Then:

- (i) There is a ring homomorphism $\psi : R \rightarrow F$;
- (ii) Any ψ as in (i) gives a bijection between the set Z of roots of f in K to the set Z_p of roots of f_p in F ;
- (iii) A function $\phi : R \rightarrow F$ is a ring homomorphism if and only if there exists $\sigma \in \text{Aut}_{\mathbb{Q}} K$ such that $\phi = \psi \circ \sigma$;
- (iv) There exists $\sigma \in \text{Aut}_{\mathbb{Q}} K$ such that $\text{Fr}_p \circ \psi = \psi \circ \sigma$.

Note that, if f is separable, then a prime p as in the statement of the theorem always exists: Since f, f' are relatively prime, there exist polynomials $g, h \in \mathbb{Q}[x]$ with $fg + f'h = 1$; then clearing denominators, we can assume $g, h \in \mathbb{Z}[x]$ and $fg + f'h = n \in \mathbb{Z}$ for some $n \neq 0$.

Then taking p to not be a factor of this n , we get that f, f' are coprime modulo p , so that f is separable modulo p .

The last statement of the theorem says that, under the homomorphism ψ , the Frobenius automorphism Fr_p lifts to an automorphism $\sigma \in \text{Aut}_{\mathbb{Q}} K$, in the sense of having a commutative diagram

$$\begin{array}{ccc} K & \xrightarrow{\sigma} & K \\ \uparrow & & \uparrow \\ R & \xrightarrow{\sigma|_R} & R \\ \downarrow & & \downarrow \\ F & \xrightarrow{\text{Fr}_p} & F \end{array} \quad (8.4.2)$$

Since Fr_p generates $\text{Aut}_{\mathbb{F}_p} F$, this immediately implies Theorem 8.1.8. (Or we could directly apply (iii), which gives the same diagram as (8.4.2) for Fr_p replaced by an arbitrary element of $\text{Aut}_{\mathbb{F}_p} F$.)

Proof of Theorem 8.4.1. First, to prove (i), note that R is a ring of algebraic integers by Propositions 8.3.2 and 8.3.5. As a result, it does not contain $\frac{1}{p}$, which has minimal polynomial $px - 1$ and is hence not an algebraic integer. So the ideal $(p) \subseteq R$ is not all of R . By commutative algebra there exists a maximal ideal $\mathfrak{m} \subseteq R$ containing p .¹ Now the quotient $R/\mathfrak{m}$ is a field of characteristic p . The image of λ_i are roots of f_p , so we get that f_p splits linearly in $R/\mathfrak{m}$. On the other hand, $R/\mathfrak{m}$ is generated by these roots. So it is a splitting field of f_p , hence isomorphic to F . This proves the assertion.

(ii) Any ψ definitely maps Z onto Z_p , because the linear factorisation of f maps to one for f_p . Since f and f_p are both separable of the same degree, it induces a bijection $Z \rightarrow Z_p$.

Before we prove (iii) we need two claims: Claim 1: R is a finitely-generated free $\mathbb{Z}$ -module. Let $\deg f = n$. Then as R is generated by the roots λ_i , it is actually spanned by $\lambda_1^{k_1} \cdots \lambda_n^{k_n}$ for $0 \leq k_i \leq n-1$ (using the fact that the λ_i all satisfy $f(x)$). So R is a finitely-generated $\mathbb{Z}$ -module. It is also free since R is a subring of K which is a field, so that R is an integral domain, hence $\mathbb{Z}$ -torsion free, then we apply the classification of finitely-generated $\mathbb{Z}$ -modules (or the fact that $\mathbb{Z}$ is a PID and the classification of finitely-generated modules over a PID). This proves the claim.

Claim 2: Let $u_1, \dots, u_d$ be a basis of R as a $\mathbb{Z}$ -module. We claim that this is also a basis of K as a $\mathbb{Q}$ -vector space. Indeed, first we see that $u_1, \dots, u_d$ is a $\mathbb{Q}$ -basis for $\mathbb{Q}[\lambda_1, \dots, \lambda_n]$: they certainly span, but are also linearly independent by clearing denominators and applying their $\mathbb{Z}$ -linear independence. Next, $\mathbb{Q}[\lambda_1, \dots, \lambda_n] \subseteq K$ is a subring and hence a field, as we have shown (in general, for $K \subseteq L$ a finite extension, and $b \in L$, then $K[b]$ is already a field since it is isomorphic to $K[x]/(g(x))$ for g the minimal polynomial of b). Since K is generated by the λ_i , we get $\mathbb{Q}[\lambda_1, \dots, \lambda_n] = K$ and the desired claim.

Proof of (iii): The group $G := \text{Aut}_{\mathbb{Q}} K$ acts on R by ring automorphisms. So for all $\sigma \in G$, it follows that $\psi \circ \sigma : R \rightarrow F$ is a ring homomorphism. Fix ψ and consider $G = \{\sigma_1, \dots, \sigma_d\}$,

¹Normally this result requires Zorn's Lemma, but in this instance it is not needed, because $R/(p)$ is actually finite, since by claim (1) in part (ii), $R/(p)$ is a finite-dimensional vector space over $\mathbb{Z}/p\mathbb{Z}$.

and take the homomorphisms $\psi_i := \psi \circ \sigma_i : R \rightarrow F$ for $1 \leq i \leq d$. These are all distinct on the roots of f by part (ii) and the fact that G is determined by its permutation of the roots ($G \leq \text{Perm}(Z)$), so that distinct σ_i induce distinct permutations of the roots.

Suppose then that $\psi_{d+1} : R \rightarrow F$ is another homomorphism. Using our $\mathbb{Z}$ -basis $u_1, \dots, u_d$ of R , we can write d linear equations for $\lambda_1, \dots, \lambda_{d+1} \in F$ which must therefore have a solution:

$$\forall j, \sum_{i=1}^{d+1} \lambda_i \psi_i(u_j) = 0.$$

If ψ_{d+1} is distinct from $\psi_1, \dots, \psi_d$, this contradicts Dedekind's independence theorem 8.2.6. So $\psi_1, \dots, \psi_d$ are all the ring homomorphisms $R \rightarrow F$, as desired.

(iv) This is merely the special case of (ii) where $\phi = \text{Fr}_p \circ \psi$. $\square$

Example 8.4.3. Let us show that the Galois group of the splitting field over $\mathbb{Q}$ of $f(x) = x^6 - 12x^4 + 15x^3 - 6x^2 + 15x + 12$ is S_6 . To see this, modulo 2 we get the factorisation $x(x^5 + x^2 + 1)$, where the second factor is irreducible since it has no roots over $\mathbb{F}_2$, and the only irreducible quadratic is $x^2 + x + 1$ which is not a factor. At the prime 3 we can apply Eisenstein's criterion to get that the polynomial is irreducible. Finally, at the prime 5, we get many roots, and get a factorisation $f(x) \equiv (x+1)(x+2)(x+3)(x+4)(x^2+3) \pmod{5}$ into irreducibles. So via Corollary 8.1.9, we get that the Galois group contains elements of cycle types (6), (5)(1), and (1)(1)(1)(1)(2), i.e., it contains a six-cycle, a five-cycle, and a two-cycle. Since it contains a six-cycle, it acts transitively on $\{1, \dots, 6\}$. That the group is S_n is a consequence of the following lemma.

Lemma 8.4.4. Let $G \leq S_n$ be a transitive group containing a transposition and an $(n-1)$ -cycle. Then $G = S_n$.

Proof. If the transposition and the $(n-1)$ cycle only overlap at one number, then up to relabeling we have (12), $(23 \cdots n)$. Multiplying these we also get (12) and $(12 \cdots n)$, which are well-known to generate (and this is easy enough to see directly). Suppose then that up to relabeling they have the form $c_2 = (2m), c_{n-1} = (23 \cdots n)$, for $m \neq n$. Since G is transitive, it also contains some g with $g(2) = 1$. Then gc_2g^{-1} has the form $(1m')$ for some m' . Now we are back in the first situation. $\square$

9 Cyclotomic polynomials

The n -th cyclotomic polynomial is the polynomial whose roots are the primitive n -th roots of unity. Normally it is defined over $\mathbb{Q}$, although really it could make sense over any field of characteristic not dividing n , since as explained in Lemma 3.8.3, in this case the splitting field of $x^n - 1$ over a field K contains a primitive n -th root of unity. It is clear that primitive n -th roots of unity are preserved under all automorphisms, so that by this definition the cyclotomic polynomial exists and is a factor of $x^n - 1$ over any field of characteristic not dividing n .

Definition 9.0.1. Let $K \subseteq L$ be a field extension such that L contains a primitive n -th root of unity, ζ . Then

$$\Phi_n(x) := \prod_{k \in (\mathbb{Z}/n\mathbb{Z})^\times} (x - \zeta^k) \in L[x]. \quad (9.0.2)$$

Note here that the expression ζ^k makes sense for $k \in \mathbb{Z}/n\mathbb{Z}$ (hence for $(\mathbb{Z}/n\mathbb{Z})^\times \subset \mathbb{Z}/n\mathbb{Z}$) as $\zeta^n = 1$.

Lemma 9.0.3. The polynomial $\Phi_n(x)$ does not depend on the choice of ζ , and lies in $K[x]$. Finally, if K is the fraction field of a UFD R , then $\Phi_n(x) \in R[x]$.

Proof. First, we can replace L with $K(\zeta) \subseteq L$ without any harm. This is a finite extension. Since we must have characteristic not dividing n , we also know that $x^n - 1$ is separable, so that $L = K(\zeta)$ is a Galois extension of K . Next, the set of all n -th roots of unity can have size at most n , but contains $\langle \zeta \rangle$, so must equal $\langle \zeta \rangle \cong C_n$. The primitive n -th roots of unity are therefore precisely ζ^k for $k \in (\mathbb{Z}/n\mathbb{Z})^\times$. So Φ is the unique monic polynomial with the primitive n -th roots of unity as its roots, and does not depend on ζ . Next, primitive n -th roots of unity are preserved by all automorphisms. So Φ_n is invariant under the Galois group $\text{Aut}_K L$. By the fundamental theorem of Galois theory, it lies in $K[x]$. For the final statement, we apply Gauss's Lemma, since $\Phi_n(x)$ is a monic polynomial which divides the monic polynomial $x^n - 1 \in R[x]$: this follows from Lemma 8.3.3. $\square$

In the case of $K = \mathbb{Q}$ that we will consider below, we may take $\zeta = e^{2\pi i/n} \in \mathbb{C}$. Recall that the group of units of a field K is denoted $\mu_n(K)$. Also recall Lemma 3.8.3, which says that the splitting field of $x^n - 1$ has a primitive root of unity if and only if the characteristic does not divide n .

Proposition 9.0.4. Suppose that $\text{char } K \nmid n$. Let L be a splitting field of Φ_n . Then the following statements are equivalent:

- (i) The cyclotomic polynomial $\Phi_n \in K[x]$ is irreducible;
- (ii) $[L : K] = |(\mathbb{Z}/n\mathbb{Z})^\times|$;
- (iii) The canonical injective homomorphism $\rho : \text{Aut}_K L \rightarrow \text{Aut } \mu_n(L) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ is an isomorphism.

Note that this statement is a more precise version of Lemma 3.8.10 in the case $\text{char } K \nmid n$.

Proof. As we noted above, L has a primitive n -th root of unity ζ , so that $L = K(\zeta)$ and $\mu_n(L) = \langle \zeta \rangle \cong C_n$. In particular, $[L : K]$ is the degree of the minimal polynomial of ζ . Since the minimal polynomial divides $\deg \Phi_n$, they are equal if and only if Φ_n is irreducible. Since $\deg \Phi_n = |(\mathbb{Z}/n\mathbb{Z})^\times|$, we get that (i) and (ii) are equivalent.

Since $L = K(\zeta)$, the homomorphism in (iii) is injective. Again by Lemma 3.8.3, L is separable, so it is Galois. By the fundamental theorem of Galois theory, $|\text{Aut}_K L| = [L : K]$. Finally, $\mu_n(L) \cong C_n$, and we have an isomorphism $(\mathbb{Z}/n\mathbb{Z})^\times \rightarrow \text{Aut}_{\text{group}} C_n$, sending m to

the automorphism $g \mapsto g^m$ (any automorphism is determined by the image of g , which can be any generator). Since an injection of finite sets is an isomorphism if and only if the sizes of the source and target are equal, (iii) is equivalent to (ii). (We could also show that (iii) is equivalent to (i) directly because $\text{Aut}_K L$ acts transitively on the roots of Φ_n if and only if Φ_n is irreducible, thanks to Proposition 5.1.2.) $\square$

We now specialise to the case $R = \mathbb{Z}, K = \mathbb{Q}, L = \mathbb{Q}(\zeta)$ for ζ a primitive n -th root of unity. The goal is to prove that $\Phi_n \in \mathbb{Z}[x]$ is irreducible, which by Gauss's Lemma is equivalent to being irreducible in $\mathbb{Q}[x]$ (as we frequently use).

Theorem 9.0.5. The conditions (i), (ii), and (iii) of Proposition 9.0.4 hold for $K = \mathbb{Q}$.

This is an amazingly deep result, with an amazingly simple proof thanks to Frobenius lifting.

Proof. We prove (iii), using Frobenius lifts. The idea is to find enough elements of $\text{Aut}_{\mathbb{Q}} \mathbb{Q}(\zeta)$ to generate $(\mathbb{Z}/n\mathbb{Z})^\times$. Let $p \nmid n$. Then $x^n - 1$ is separable in $\mathbb{F}_p$. Let F be its splitting field. Thanks to Theorem 8.4.1.(i), there is a ring homomorphism $\psi : \mathbb{Z}[\zeta] \rightarrow F$ inducing a bijection on n -th roots of unity. Moreover, by (iv), there exists an element $\sigma \in \text{Aut}_{\mathbb{Q}} \mathbb{Q}(\zeta)$ which lifts the Frobenius map on roots of n -th unity, $\psi \circ \sigma = \sigma \circ \text{Fr}_p$. But the Frobenius map on $\mu_n(F)$ is of the form $z \mapsto z^p$. Since ψ is a homomorphism, we get $\sigma(\zeta) = \zeta^p$. So we get that the image of the canonical map $\text{Aut}_{\mathbb{Q}} \mathbb{Q}(\zeta) \rightarrow \text{Aut}_{\mu_n(K(\zeta))}$ includes $\zeta \mapsto \zeta^p$, which is identified with $p \in (\mathbb{Z}/n\mathbb{Z})^\times$.

Now the claim is reduced to: the primes not dividing n generate $(\mathbb{Z}/n\mathbb{Z})^\times$. But take any $m \in (\mathbb{Z}/n\mathbb{Z})^\times$. We can write it as a positive integer coprime to n . Write the prime factorisation of m . Then it is clear that it is generated by the primes coprime to n (in fact, we only need the primes less than n). $\square$

Remark 9.0.6. Note that Corollary 8.1.9 of Frobenius lifting we used before is useless here, since we don't know how $x^n - 1$ factors modulo p —in fact, what we want to show is that Φ_n is an irreducible factor, and if we knew that was true modulo p then it would automatically be true over $\mathbb{Q}$. Instead, we really used an explicit formula for the Frobenius lift.

Remark 9.0.7. Caution: This explicit formula for the Frobenius lift, $\zeta \mapsto \zeta^p$, was deceptively simple, working because roots of Φ_n are closed under taking p -th powers. For general $f \in \mathbb{Z}[x]$ with separable reduction modulo p , the lift σ of the Frobenius will still be determined by the condition $\psi(\sigma(\alpha)) = \psi(\alpha)^p$, but usually $\sigma(\alpha) \neq \alpha^p$ since α^p won't be a root of f , so finding $\sigma(\alpha)$ may not be easy.

Example 9.0.8. Let's compute the Galois group of $x^{12} - 2$. This is generated by a primitive twelfth root of unity, ζ , and $\sqrt[12]{2}$. Let $K := \mathbb{Q}(\zeta)$, we have shown above that $\text{Aut}_{\mathbb{Q}} K \cong (\mathbb{Z}/12\mathbb{Z})^\times \cong (\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z})^\times \cong C_2 \times C_2$. We claim that $x^{12} - 2$ is irreducible over K . Assuming this claim, we have that its splitting field $L := K(\sqrt[12]{2})$ satisfies $\text{Aut}_K L \cong C_{12}$ by Proposition 3.8.4.

We have $\text{Aut}_{\mathbb{Q}} L / \text{Aut}_K L \cong \text{Aut}_{\mathbb{Q}} K$. Thus $|\text{Aut}_{\mathbb{Q}} K| = 12|(\mathbb{Z}/12\mathbb{Z})^{\times}| = 48$. Automorphisms are uniquely determined by sending ζ to any other primitive twelfth root, ζ^{ℓ} for $\ell \in (\mathbb{Z}/12\mathbb{Z})^{\times} \cong C_4$, and $\sqrt[12]{2}$ to $\zeta^m \sqrt[12]{2}$ for $0 \leq m \leq 11$, independently.

We can thus take a group homomorphism $(\mathbb{Z}/12\mathbb{Z})^{\times} \cong \text{Aut}_{\mathbb{Q}} K \rightarrow \text{Aut}_{\mathbb{Q}} L$ sending $m \in (\mathbb{Z}/12\mathbb{Z})^{\times}$ to the map $\zeta \mapsto \zeta$, $\sqrt[12]{2}$ to $\zeta^m \sqrt[12]{2}$. Put together, we get a splitting of the quotient $\text{Aut}_{\mathbb{Q}} L / C_{12} \cong (C_2 \times C_2)$, and deduce $\text{Aut}_{\mathbb{Q}} L \cong (C_2 \times C_2) \rtimes C_{12}$, otherwise written as $(\mathbb{Z}/12\mathbb{Z})^{\times} \rtimes \mathbb{Z}/12\mathbb{Z}$.

In the coursework you will be asked to generalise this to arbitrary $x^n - a \in \mathbb{Q}[x]$ for arbitrary n and a . Isn't that exciting?

10 Kummer theory

As a final topic, we would like to revisit the theory of cyclic Galois extensions initiated in Section 3.8. Let us first summarise what we already proved there. Let K be a field of characteristic p (which could be zero).

- The polynomial $x^n - 1 \in K[x]$ is separable if and only if $p \nmid n$;
- The splitting field of $x^n - 1$ contains a primitive n -th root of unity if and only if $p \nmid n$;
- If K contains a primitive n -th root of unity, then the splitting field of $x^n - a$ is a Galois extension with cyclic Galois group of order dividing n ;
- If $K = K_0 \subseteq K_1 \subseteq \dots \subseteq K_k = L$ are field extensions with $K_{i+1} \supseteq K_i$ given by adjoining an n_i -th root of an element of K_i where K contains an n_i -th primitive root of unity, then L is a Galois extension of K with solvable Galois group.

Moreover, given ζ a primitive n -th root of unity, $K \subseteq K(\zeta)$ is a Galois extension with abelian Galois group a subgroup of $(\mathbb{Z}/n\mathbb{Z})^{\times}$ (this is Lemma 3.8.10, not covered back then, but will be covered now). As a result, we can remove the hypothesis of containing an n_i -th primitive root of unity in the preceding statement, by adding in field extension by primitive roots of unity, provided that $p \nmid n_i$. We summarise:

Proposition 10.0.1. If L is obtained from K by iterated extensions adjoining n_i -th roots of elements, for $\text{char } K = p \nmid n_i$, then L is a Galois extension with solvable Galois group.

In particular, this proposition shows that the polynomial $x^5 - x - 1$ in Example 8.1.10 does not have roots expressible using radicals over $\mathbb{Q}$.

The goal of this section is to add converses to the above statements, producing the following results, under the heading of “Kummer theory”:

Theorem 10.0.2. Suppose that K contains a primitive n -th root of unity. The following are equivalent:

1. $L = K(b)$ where $b^n \in K$, $b^m \notin K$ for $1 \leq m < n$;

2. L is a Galois extension of K with cyclic Galois group of order n .

Example 10.0.3. Here is a simple example to show that the primitive n -th root of unity is needed for (1) implies (2): the field $L := \mathbb{Q}(\sqrt[3]{2})$ is not Galois, as the minimal polynomial of $\sqrt[3]{2}$ is $x^3 - 2$ but the other two roots of this are complex. If we instead consider the splitting field of $x^3 - 2$, as we have seen we get S_3 , which is solvable but not cyclic.

Example 10.0.4. For (2) implies (1), we also need the primitive n -th root of unity. In the biquadratic setting, we had the extension $L = \mathbb{Q}(\sqrt{2 + \sqrt{2}})$ of $K = \mathbb{Q}$ with Galois group C_4 . Here K and even L contains no primitive fourth root of unity, $\pm i$, as L is real. But L can't be given from K by taking a fourth root, since any real such extension can't be normal.

Example 10.0.5. In positive characteristic stranger counterexamples can happen without the primitive root of unity. For example, let $K = \mathbb{F}_2$ and let L be the splitting field of $x^2 + x + 1$. We have $L \cong \mathbb{F}_4$, a Galois extension with group C_2 . This cannot be given by taking square roots as that can only produce inseparable extensions, but there is also no primitive square root of unity.

Note in this example that $x^2 + x + 1$ is the cyclotomic polynomial Φ_3 , so $L = K(\omega)$ for $\omega^3 = 1$ a primitive cube root of unity. So we also see that adjoining primitive n -th roots of unity can produce cyclic Galois extensions of degree a multiple of the characteristic, in general.

Iterating the theorem, we obtain:

Corollary 10.0.6. Suppose that K contains a primitive n -th root of unity. Then the following are equivalent:

1. There is a chain $K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n = L$ where each $K_i \subseteq K_{i+1}$ is given by adjoining an n_i -th root for $n_i \mid n$;
2. L is a Galois extension with solvable Galois group of order dividing n^k for some $k \geq 1$.

Proof. First note that we don't need to say anymore that the extension is not given by a smaller than n_i -th root, since in part (2) also we only require the order to divide n^k . Finally, to have a primitive n -th root of unity, we must have $\text{char } K \nmid n$, so it's still true that the last statement implies that the order is not a multiple of $\text{char } K$. $\square$

Stating this in terms of splitting fields of polynomials, we get:

Corollary 10.0.7. Now let $f \in K[x]$. Suppose that K contains a primitive n -th root of unity. Then the following are equivalent:

1. The polynomial f is solvable by radicals of order dividing n ;
2. The splitting field of f is a Galois extension with solvable Galois group of order dividing n^k for some $k \geq 1$.

All that is left is to prove Theorem 10.0.2.

Proof of Theorem 10.0.2. Let $\zeta \in K$ be the primitive n -th root of unity assumed.

First, assume (1). We claim that $x^n - b^n$ is irreducible. Given this, we obtain (2) from Lemma 3.8.3 and Proposition 3.8.4 (the source of the facts listed at the beginning of the section). To prove that $x^n - b^n$ is irreducible, note that every monic factor has the form $(x - \zeta^{i_1}b) \cdots (x - \zeta^{i_k}b)$ for some $i_1, \dots, i_k$. If this has coefficients in K , then the constant term yields $\zeta^{i_1 + \dots + i_k} b^k \in K$. Since $\zeta \in K$, we have $b^k \in K$. Now for some $m \in \mathbb{Z}$ we have $0 \leq k - mn < n$. By assumption this implies that $0 = k - mn$, i.e., $k = mn$, so in this case $k = n$ or $k = 0$, and the only factors of $x^n - b^n$ in $K[x]$ are 1 and the whole polynomial (up to constant factor). This proves the claim.

The main step is to prove that (2) implies (1). Suppose that $K \subseteq L$ is a Galois extension with Galois group C_n . Let σ generate C_n . The goal is to find $b \in L$ nonzero such that $\sigma(b) = \zeta b$. We will prove such a b exists below. Given this, then b^k is an eigenvalue of σ of eigenvalue ζ^k for all k . By linear algebra, since $1, \zeta, \dots, \zeta^{n-1}$ are distinct eigenvalues, then $1, b, b^2, \dots, b^{n-1}$ are K -linearly independent elements of L . By Galois theory, $[L : K] = n$, so we find that $1, b, \dots, b^{n-1}$ is a K -linear basis of L , i.e., $L = K(b)$. But, $\sigma(b^n) = \zeta^n b^n = b^n$, so $b^n \in L^\sigma = L^{\text{Aut}_K L} = K$, so that (1) holds.

To find b , we apply Dedekind's independence theorem 8.2.6 to the characters $1, \sigma, \dots, \sigma^{n-1} : L \rightarrow L$. The theorem yields that these are L -linearly independent. So the linear combination $\sum_{i=0}^{k-1} \zeta^{-i} \sigma^i$ is not identically zero, and in particular there exists $x \in L$ such that

$$b := \sum_{i=0}^{k-1} \zeta^{-i} \sigma^i(x) \in L$$

is nonzero. Since $\zeta^{-n} \sigma^n = 1$, we see that $\zeta^{-1} \sigma(b) = b$, i.e., $\sigma(b) = \zeta b$, as desired. $\square$